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TAXING ENTREPRENEURIAL WEALTH: EVIDENCE FROM NORWAY, 2021–2025

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ABSTRACT

Whether imposing higher taxes on business owners adversely affects business activity by constraining investment and inducing capitalist flight is a central question in the ongoing debate on how to tax the ultra wealthy. To shed new light, I exploit a series of related Norwegian reforms during 2021–2024 that increased dividend tax rates, removed migration-related capital gains tax loopholes, and nearly doubled the effective marginal tax rate on business wealth. These reforms spurred international debate and, allegedly, an "exodus" of Norwegian billionaires (Financial Times, 2023). While I document clear effects on migration, these responses are concentrated among the top 0.1% of the wealth distribution and mostly confined to 2022 and 2023. My results indicate that at most 100 individuals left due to the reforms. I find no evidence of significant outmigration or reduced inflows of startup founders or inventors. I estimate that absent any outmigration, wealth tax revenues would have increased by 75%from 2021 to 2024. Reform-driven outmigration lowers the revenue gain to 71.5%. My findings further show that outmigrating owners' firms remain economically active and do not reduce investment. In a broader set of analyses, I examine whether business outcomes are affected, regardless of whether owners outmigrate, since firms may reduce investment both due to liquidity constraints or distortions to owners' savings decisions. I compare firms owned by Norwegian individuals subject to the wealth tax with firms owned by foreigners and other tax-exempt entities. By merging historical ownership data from Orbis with the Norwegian shareholder register, I perform these analyses on both Norwegian and foreign-domiciled firms. Across settings, I find no evidence that Norwegian-owned firms pay out more dividends or invest less after the reforms, nor is there any evidence that the reform induced more bankruptcies. These findings indicate that the short-run economic costs of wealth taxation, in terms of migration and firm investment, are more limited than the public debate implies.

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1 Introduction

The past decade has seen renewed interest in wealth taxation as a tool to raise tax revenues and address economic inequality. The motivation for more aggressive taxation of wealthy individuals is partly driven by the modest personal tax burdens faced by billionaires. A handful of recent studies show that once undistributed profits are accounted for, there is significant tax *regressivity* at the very top of the income distribution (Bruil, Van Essen, Leenders, Lejour, Möhlmann, and Rabaté 2022; Aaberge, Francesconi, Modalsli, and Vestad 2024; Balkir, Saez, Yagan, and Zucman 2025; Bach, Bozio, Guillouzouic, and Malgouyres 2025; Ring, Seim, and Zucman 2026). The wealthiest individuals tend to be business owners that retain most of their profits inside closely-held corporations, rarely realize capital gains (Msall and Næss, 2026), and thereby avoid personal income taxation. A wealth tax (or similarly, a tax on unrealized capital gains) could remedy this regressivity in that it does not condition on corporate distributions. However, the fact that wealthy individuals retain earnings in their firms also fuels the vocal opposition to wealth taxation. Imposing large personal tax bills on billionaire entrepreneurs may force them to deplete their firms of capital, potentially reducing investment and thus economic growth. This concern pervades policy debates despite virtually no supporting empirical evidence.

There are two main drivers for why there is little empirical evidence to guide policy makers. First, most recent experiences with a wealth tax involve abolishing it. Hence, there are few empirical laboratories in which we can study how introducing (more) wealth taxation affects economic activity. Introducing a wealth tax may reduce economic activity through irreversible firm exits and reductions in investment that cannot be fully understood by studying the removal of wealth taxes. Second, identifying the causal effects of wealth taxes on wealthy entrepreneurs is particularly challenging. While it is possible to obtain quasi-experimental variation in wealth taxation due to quirks in the tax code, these settings rarely allow one to study the wealthiest households. For these taxpayers, whether couples face a double exemption threshold (Jakobsen et al. 2020) or their housing stock is assessed more heavily (Ring 2025, Ring and Thoresen 2025) is immaterial in affecting their overall tax burden.

I tackle these empirical challenges in two ways. First, my empirical laboratory consists of a recent reform in Norway that substantially *increased* the tax burden faced by the wealthiest households. Second, I identify the effects on firms by exploiting the fact that the wealth tax only applies to Norwegian tax residents. Hence, firms owned by non-resident foreigners should not be affected—offering a natural control group for the firms of wealthy Norwegians.[1] This allows me to control for a wide range of firm characteristics, such as firm size, without

[1] See also Le Guern Herry (2024) who exploits resident and non-resident taxpayers' differential exposure to a wealth tax reform in France to study household portfolio allocation.

removing the identifying variation. This is not possible in settings in which researchers compare wealthy individuals to less exposed, less wealthy individuals, since less wealthy entrepreneurs own smaller firms.

The reforms implemented during 2021–2024 were controversial. Upon announcing higher wealth and dividend tax rates in 2021, a political and media frenzy quickly ensued. The public debate persisted well into 2025, by which time scores of billionaires had, not so quietly, migrated to Switzerland (Financial Times 2023, Bloomberg 2024, Guardian 2025). Migrants purportedly left due to several changes to the wealth tax code that particularly affected entrepreneurs. The top statutory tax rate increased from 0.85% to 1.1% while preferential treatment of stocks in private and public companies was substantially reduced. Directly before the reform, each \$1 of book equity in private firms entered taxable wealth at a discounted value of \$0.55. After the reform, \$1 of book equity increased taxable wealth by \$0.80. Hence, the reform almost doubled the top marginal tax rate on business wealth from 0.47% to 0.88%. I show that this had a material impact on the annual wealth tax paid by Norwegian business owners. At the same time, the dividend tax rate increased from 32% to almost 38% by 2023, which exacerbated the effect of the reform on business owners who rely on dividend distributions to pay their wealth tax.

The government responded to the widely publicized outmigration of wealthy business owners by implementing new policies to ensure that outmigrants eventually paid income taxes on accumulated retained earnings. In November of 2022, the government removed a loophole that forgave tax liabilities on unrealized gains after a taxpayer had spent at least five years as a foreign tax resident. Closing this loophole, but allowing taxes on unrealized capital gains to be permanently deferred brought the rules closer to the current scheme for outmigrants in Denmark. This policy change was followed by stricter rules in 2024, when it was announced that taxes on unrealized capital gains must be paid within 12 years, bringing the exit-tax scheme closer to current rules in, e.g., Germany.

This paper pursues two lines of analysis. I study whether the 2021–2024 reforms led to capitalist flight and the extent to which firms owned by affected Norwegian business owners increased their dividends and reduced their investment rates. I use a broad mix of publicly available and restricted-access data sources. These include the shareholder register that lists all shareholders and their ownership shares for all Norwegian corporations during 2003–2025. These data also contain the address (including country of residence) of the shareholders. I link these with extensive public tax lists containing taxable wealth, income, and total taxes paid for wealthy Norwegians, and with journalistic estimates of actual wealth for the wealthiest 400 Norwegians. I obtain firm financial data from BvD Orbis Historical, which provides a long panel of financial statements for Norwegian firms. Orbis Historical also contains financial statements

for a large sample of other European firms, as well as ownership data. These ownership data allow me to link non–Norwegian firms to their ultimate Norwegian owners by merging the BvD ownership data with the Norwegian shareholder register.

The most salient and widely reported effect of the reforms was the departure of wealthy individuals to Switzerland. Consistent with this, Blandhol (2025) documents significant excess outmigration at the very top of the Norwegian wealth distribution in 2022. My analyses extend the time horizon to evaluate whether outmigration was a transitory response to pre-empt the closure of tax loopholes or persistent responses to wealth taxation. I also examine migration responses and their dynamics in several policy-relevant subpopulations. I first use administrative data to show that there was significant excess outmigration following the reform for individuals in the top 0.1% of the taxable wealth distribution. This excess outmigration lasted to the fall of 2023, after which migration rates returned to pre-reform levels. My estimates suggest that about 1.78% of the top 0.1%, or about 0.24% of the top 1% migrated due to the reform. This amounts to 74–98 individuals. The slow-down of outmigration coincides with the closure of tax-related migration loopholes, which suggests that these loopholes were an important driver of the response. However, the closing of the five-year loophole may also have accelerated wealth-tax-motivated migration.

Since these outmigrating individuals were particularly wealthy, the revenue effects are clearly discernible. Contrasting observed aggregate tax revenues with those that would have prevailed absent any outmigration, I estimate a total wealth-tax revenue loss of about 0.65 BNOK, or about 2% of counterfactual revenues. Nevertheless, these avoidance responses do not come close to countering the strong mechanical effects of the reform, causing an increase in wealth tax revenues of about 71.5% net of reform-driven outmigration. I further find no evidence that firms of owners that outmigrated between May 2022 and May 2024 are less likely to remain active or invest less by the end of 2025.

To understand the anatomy of the response, I study outmigration responses in several subpopulations. I consider high-profile billionaires featured in the top-400 wealth ranking by the magazine *Kapital*. I construct a novel dataset that provides a high-frequency outmigration panel by merging the wealth rankings with address information from the shareholder register and newspaper mentions of migration events (e.g., "tax refugee" status). These data indicate substantial reform-induced outmigration among Norway's wealthiest individuals. Interestingly, most of this migration occurred in the fall of 2022—directly before the five-year loophole was closed. In this sample of high-profile billionaires, there is no additional outmigration during 2024 or 2025, when the exit-tax loopholes had mostly been closed. I also consider individual shareholders in Norwegian firms. I find excess outmigration of 2.19% among the top-1% of business owners, but no response among the remaining 99%.

The fact that many wealthy individuals chose to leave the country after the reform begs the question of whether this effect spread to startup entrepreneurs and innovators who may rely on external financing. To address this, I use data on the near universe of LinkedIn profiles from RevelioLabs to study migration rates of startup founders. I then contrast the migration rates of Norwegian entrepreneurs with other Nordic entrepreneurs. The profile classification system allows me to isolate "startup" and "tech founders" from the broader pool of owner-managers. And unlike standard administrative data that are typically limited to one country, the LinkedIn data allow me to study the effect on inflows of entrepreneurs from other countries. These analyses do not indicate any excess outflow or reduced inflow of entrepreneurs in Norway. I further link the LinkedIn data with patent records to study the effect on inventor mobility. Again, there is no evidence that more inventors left or fewer inventors arrived in Norway relative to other Nordic countries. This begs the question of whether startup investors, as a distinct group, actually fled Norway following the reforms. To examine this, I consider the subsample of business owners who had previously invested in innovative, grant-receiving startups. In this subsample, there is no excess outmigration. On the contrary, it appears that outmigration rates dropped below pre-period trends as migration-related tax loopholes were closed.

My next analyses concern whether the tax reforms affected firms' financing and investment behavior, regardless of whether their owners outmigrated. I first demonstrate that whether higher wealth taxes reduce corporate investment is theoretically ambiguous.[2] This result arises naturally in an entrepreneurial life-cycle model with financial frictions. When the elasticity of intertemporal substitution (EIS) is low enough, the owner may wish to save more to offset the adverse effects of wealth taxation on future consumption. In this case, wealth taxes may lower the user cost of capital. In a simple calibrated model, the 2021–2024 reforms only decrease investment if the EIS exceeds about 0.66, a value that lies between existing estimates from Norway (Ring 2025; Holm, Jamilov, Jasinski, and Nenov 2026). The implied change in the investment rate ranges from -2.8 to $+3.5$ percentage points for values of the EIS between 0.2 and 6.

My empirical approach compares firms owned by Norwegian individuals subject to the wealth tax with firms owned by exempt owners such as foreign corporations or individuals. To account for the fact that foreign-owned firms tend to be larger and operate in specific industries, I employ a rich set of granular industry and size fixed effects. To measure a firm's exposure to the reform, I allocate an owner's wealth tax liability to her firm in proportion to ownership-weighted revenues. I then consider the ratio of allocated wealth tax to revenues, and find that this increased by about 1.6 percentage points from 2020 to 2024. To investigate whether increased wealth taxation required business owners to deplete their firms of capital via

[2]Under specific conditions on outside options, investment may also be neutral to wealth taxation (see, e.g., Bjerksund and Schjelderup 2022).

higher dividends, I measure "operating dividends" as net income minus financial income and the change in retained earnings. This measure of operating dividends ignores dividends that pass through firms in complicated ownership chains by excluding financial income from net income. There is no indication that Norwegian owned firms increased their dividends during 2022–2024. I proceed to measure a firm's investment intensity as the ratio of investment (change in book tangible fixed assets plus depreciation) to its tangible-fixed-asset capital stock, according to the perpetual inventory methodology. I find no indication of reduced investment rates during 2022–2024. The main estimates are precise. I can reject reductions in the investment rate larger than 1.2 percentage points, about half the predicted reduction in a standard model with no financial frictions, in which the reforms increase the user cost of capital.

To account for potential identification issues related to comparing foreign and domestically-owned firms, I repeat the main analyses on a sample of foreign-domiciled corporations. That is, I consider European companies in the Orbis database that have global ultimate owners domiciled in other countries. I then compare, e.g., Norwegian-owned German firms with Danish-owned German firms. This confirms the null effect on investment.

A final set of analyses examine whether the reforms disrupted entrepreneurial finance, which depends on entrepreneurs' personal wealth (Chodorow-Reich, Nenov, Santos, and Simsek 2024; Ring 2023). I first examine whether the reform induced excess outmigration among startup investors, defined as individuals with minority stakes in young firms that have received innovation grants. This reveals no indication of excess outmigration. Rather, there appears to be less outmigration once migration-related tax loopholes were removed. I also indirectly examine whether wealthy Norwegians reduced their startup investments after the reform. I consider both ownership in innovation grant recipients and young firms receiving R&D credits. For innovation grant recipients, there is no indication of reduced investment from Norwegian investors. For innovation-tax-credit recipients, the data indicate reduced investment, particularly when weighting by invested capital.

Literature. A growing body of research examines the behavioral responses to wealth taxation (Seim 2017; Zoutman 2018; Bach, Bozio, Guillouzouic, and Malgouyres 2020; Jakobsen, Jakobsen, Kleven, and Zucman 2020; Berg and Hebous 2021; Londoño-Vélez and Ávila-Mahecha 2021; Brülhart, Gruber, Krapf, and Schmidheiny 2022; Alstadsæter, Bjørneby, Kopczuk, Markussen, and Røed 2022; Fagereng, Guiso, and Ring 2023; Jakurti and Süssmuth 2023; Le Guern Herry 2024; Ring 2025; Mas-Montserrat, Durán-Cabré, and Esteller-Moré 2025; Londoño-Vélez and Avila-Mahecha 2025; Ring and Thoresen 2025), where the emerging consensus is that wealth taxation has minor real effects in terms of individuals' saving behavior (as emphasized by Brülhart et al. 2022 and Ring 2025). However, the tax base may still be elastic due to evasion, avoidance, and migration.

A much smaller body of research examines mobility responses. Brülhart, Gruber, Krapf, and Schmidheiny (2022) find significant elasticities of taxable wealth with respect to marginal tax rates in Switzerland, of which one quarter are driven by within-country but across-tax-regime migration responses. Iacono and Smedsvik (2024) similarly finds that a Norwegian municipality that reduced its local tax rate saw a large increase in the municipal-level amount of taxable wealth. Agrawal, Foremny, and Martínez-Toledano (2025) further document significant within-country mobility responses in Spain. After the 2011 reintroduction of the wealth tax, which Madrid chose not to levy, the wealth-tax-paying population (top 0.5%) in the other regions fell by 1.7% within six years. Considering outmigration, Jakobsen, Kleven, Kolsrud, Landais, and Muñoz (2024) find that the 2007 abolition of the Swedish wealth tax, which averaged about 0.5% of net wealth for the top 2%, reduced their annual outmigration rate by 0.05 percentage points. For Norway, Grindaker and Vestad (2025) estimate that the emigration probability of the top 1% increased by 0.27 percentage points per percentage-point increase in the wealth tax rate, a response they note partly reflects the November 2022 tightening of the exit tax.

An even smaller body of research considers the real effects of wealth taxation on the firms owned and financed by wealth tax payers. The existing evidence is inconclusive and identified from gradual changes to the wealth tax rules. Bjørneby, Markussen, and Røed (2023) find that higher wealth tax exposure increases employment while Berzins, Bøhren, and Stacescu (2020) find the opposite.[3]

The paper proceeds as follows. Section 2 describes the 2021–2024 reforms and the data. Section 3 studies outmigration and tax revenue effects. Section 4 presents a stylized user-cost model and then estimates the effects of the reforms on firms' payouts, investment, and bankruptcy. Section 5 examines potential disruptions to entrepreneurial finance. Section 6 concludes.

2 Institutional Details and Data Sources

2.1 The 2021–2024 Reforms

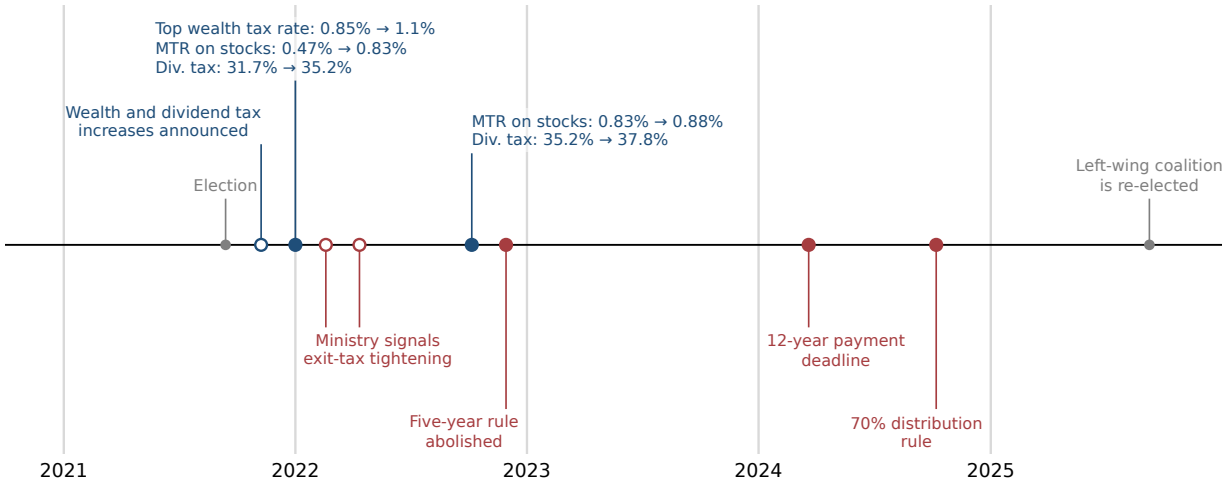
The wealth tax. Norway levies an annual progressive tax on net wealth. The relevant tax base is taxable net wealth, consisting of financial and real assets net of debt. As of 2021, any taxable net wealth in excess of 1.5 MNOK ($150,000) is subject to a 0.85% tax. Listed stocks enter at a 45% discounted market value. Unlisted stocks enter approximately at the

[3] Bjørneby, Markussen, and Røed (2023) obtain variation in wealth tax exposure from tax schedule changes that differentially affected households with different taxable wealth composition. Similarly, Berzins, Bøhren, and Stacescu (2020) compare individuals with more or less housing wealth that were differentially affected by gradual increases in the effective taxation of housing wealth during 2006–2009.

book-value of equity after the 45% discount is applied. The tax authorities do not make use of implied valuations from external financing (e.g., VC funding) to assess the value of a private firm. Holdings in other financial securities, such as bank deposits and bonds, enter at their market values. The value of primary and secondary residences is estimated using a hedonic pricing model (see Ring (2025) for details). Primary homes are subject to a 75% discount and secondary homes are subject to a 10% discount. Third-party reporting is ubiquitous: stock ownership is reported by the issuer or financial intermediaries, the value of other securities or savings accounts is reported by financial institutions, and homeownership is established via real-estate registers.

The 2021 election. The election replaced the previous right-wing coalition government (2013–2021) with a left-wing coalition government. By 2022, the left-wing government had implemented several changes to the tax code that affected (wealthy) business owners. Figure 1 summarizes the timing of the reforms.

FIGURE 1: TIMELINE



Partially reversed the assessment discounts on stocks. The raw tax base for listed stocks is the market value of the shares. For non-listed stocks, the raw tax base is the book value of equity subject to various adjustments (e.g., if the corporation owns listed stocks), which is typically below market value (Andresen and Eide Bø 2022; Grindaker and Vestad 2025; Skar, Traini, Schmidt, and Solheim 2025). In 2017, the right-wing government began phasing in an assessment discount which created a wedge between the raw tax base and the value used to determine the wealth tax. In 2017, the discount rate was 10%, and in the following years it grew to 20%, 25%, and 35%. By 2021, the discount had reached 45%, which implied that

each additional $1 of book-value business wealth increased taxable wealth by only $0.55. The left-wing government decreased this discount to 25% in 2022 and 20% in 2023, where it remains as of 2026.[4]

Increased statutory tax rate and thresholds. The new government modestly increased the wealth-tax exemption threshold from 1.5 MNOK in 2021 to 1.7 MNOK in 2022. At the same time they increased the baseline tax rate (0.85% during 2015–2021) to 0.95% in 2022 and 1.0% in 2023. The left-wing government also increased the progressivity of the tax schedule by adding another threshold of 20 MNOK, above which any marginal wealth is subject to an additional 0.15 percentage points in 2022 and 0.10 percentage points from 2023. Thus the top marginal tax rate increased from 0.85% in 2021 to 1.1% in 2022.[5]

For wealthy business owners, the combination of a lower assessment discount on stocks and higher marginal tax rates led to a sharp rise in the marginal tax rate on business wealth, from $0.85\% \times 0.55 = 0.4675\%$ in 2021 to $1.1\% \times 0.8 = 0.88\%$ in 2023.

Dividend taxation. The dividend tax rate was 28.75% in 2016, and rose gradually to 31.68% by 2019.[6] The dividend tax rate then stayed at this level until the new government increased it to 35.20% in 2022, and to 37.84% in 2023. The increase to 35.20% was announced by the new government roughly seven weeks ahead of time, giving entrepreneurs the opportunity to pay more dividends before the higher rate became effective, as they did ahead of the 2006 and 2016 increases (Alstadsæter, Kopczuk, and Telle 2014; Andresen 2021).[7]

Exit taxation. Prior to November 2022, individuals would see their stocks effectively "step up" in basis five years after outmigration.[8] They would then be free to sell their shares without paying capital gains taxes, even if this occurs after returning to Norway. In February 2022, the Minister of Finance indicated publicly, through a written and public Q&A correspondence with a member of parliament, that the Ministry was exploring options to tighten these rules (Stortinget, 2022b). By April 2022, the government confirmed to the finance committee that this work was already under way (Stortinget, 2022a). On November 29th, 2022, the government announced that individuals who migrate on or after that date can no longer make use of the

[4]The new government's budget proposal of November 8th, 2021 reduced the discount only to 35%. The further reduction to 25% was settled in the budget agreement of November 29th, 2021.

[5]This brought the top marginal tax rate back to same level as during 1998–2013.

[6]During 2010–2013, the tax rate was 28%. During 2014–2015, it was 27%.

[7]The further increase to 37.84% was given effect from the day the 2023 budget was presented, October 6th, 2022. The stated rationale was that earlier increases had produced large distributions ahead of the effective date.

[8]Norway abolished step-up in basis at inheritance for stocks in 2006 and for real-estate in 2014, implying that heirs carry over the basis (Msall and Næss, 2026). After 2014, the five-year emigration loophole was the most straightforward way for wealthy dynasties to be able to realize gains without paying income taxes.

five-year exemption rule and, in principle, will be liable for taxes on the amount of unrealized gains at the time of outmigration. However, there was no deadline by which the taxes need to be paid.

In March 2024, the government indicated that they would implement a payment deadline of 12 years (Finansdepartementet, 2024a). This change moved the rules closer to those implemented in Germany in 2021, where taxes on unrealized capital gains need to be paid within 7 years of outmigration. In comparison to the U.S., the new Norwegian rules are more generous in that they do not require interest payments, but less generous in the sense that the tax must be paid within twelve years as opposed to at the actual realization event. However, even after the new exit tax rules of 2022 and March 2024, there were still legal opportunities to indefinitely avoid paying either dividend or capital gains taxes for new emigrants (Torvik et al., 2025). The exit tax was assessed at outmigration but was only triggered by sales. Hence, an emigrant could freely extract dividends without having to pay any exit tax. In principle, new emigrants faced a 15–25% dividend withholding tax (depending on tax treaties), but the withholding tax could be avoided altogether through a loophole discussed by Torvik et al. (2025). Prior to outmigration, owners could exchange their shares, tax-free, for shares in a foreign holding company that would now own the Norwegian firm (Skattedirektoratet, 2023).[9] The Norwegian company would then pay dividends to the foreign holding company, without triggering any dividend or withholding taxes. The foreign holding company could then pay the dividends to the now-foreign-resident individual at the foreign tax rate (which can be low for wealthy foreigners who elect lump-sum taxation in Switzerland). Having stripped the firm of its value, the emigrant could return to Norway within the 12-year deadline. Upon return, the exit tax liability no longer applies, and the owner would not have to pay dividend taxes on dividends received while abroad.

On October 7th, 2024, the government blocked this strip-and-return strategy with immediate effect (Finansdepartementet, 2024b, Lovdata, 2024b). For individuals emigrating on or after October 7th, any subsequent distribution on shares subject to the exit tax causes 70% of the distributed amount to fall due as payment on the exit tax liability, whether the distributing company is Norwegian or foreign.[10]

[9] Such a share exchange ("aksjebytte") is tax-free for the owner under skatteloven § 11-11, and the deferred exit tax carries over to the new shares. The holding company would need to be incorporated in an EEA state to avoid the dividend withholding tax.

[10] Skatteloven § 10-70, eighth paragraph, as amended by lov 20. desember 2024 nr. 86. Distributions include dividends, any gratuitous transfer of value to the shareholder or related parties, and shareholder loans, but not repayment of paid-in capital (Finansdepartementet, 2024b). The 70% factor is a standardized allowance for taxes on the distribution itself. There is no credit for withholding tax. The payment is reduced where taxes on the distribution plus the payment would exceed the distributed amount, that is, where dividend taxes exceed 30% of the distribution (Stortinget, 2024, Lovdata, 2024a).

The 2025 election. Despite the controversies surrounding the government's tax reforms, in particular those related to the wealth tax, the left-wing coalition again obtained a majority of the parliamentary seats in the September 2025 elections. This was not due to low election turnout among the ultra wealthy. In Appendix Figure OA.1, I document that turnout at the 2025 elections was 97.9% among the top 0.01%. The implied abstention rate of 2.1% is considerably lower than in prior parliamentary elections, where the top 0.01% abstained at a rate of 7.6% in 2021 and 5.7% in 2017.

2.2 Data

Orbis Historical. The Orbis historical data contains all financial statements for firms present in the BvD (now Moody's) Orbis dataset. The coverage for Norwegian firms is nearly complete due to Orbis purchasing these data from Experian who obtain them from the Norwegian business registration office.

Shareholder Register. The shareholder register for 2003–2025 is obtained from the tax authorities. These data contain the most recent address of shareholders as of delivery. Hence, I use several different vintages of the shareholder register (2014 through 2025) to obtain a panel of shareholder locations. The shareholder register contains the name of the shareholder and their birth year (organization number if a domestic LLC).

Administrative microdata. For some analyses, I use administrative micro data accessed via the `microdata.no` platform. This platform allows researchers to access a broad set of administrative registers via a secure interface through which one can perform statistical analyses but not extract or view the microdata themselves. I use this service to analyze migration rates across the taxable wealth distribution during 1994–2025. These data cannot be and are not linked with the other data sources. See Appendix B.3 for details on the use of this data.

Newspaper Tax Lists. Individuals' taxable income, taxable wealth, and total taxes paid are considered public information in Norway. Each year, since 2013, the tax authorities send complete "tax lists" to media organizations who may choose to disclose information on persons of interest. The tax authorities explicitly allow media organizations to publish "top-100" lists: top 100 individuals per municipality in terms of taxable personal income, taxable wealth, and taxes paid. In addition, since 2019, it is common to publish top-15 zip-code lists. I obtain all of these lists for the years 2012–2024 (zip-code lists available from 2019). These lists contain full names, birth year, and address (zip code, city) information.

Magazine wealth rankings. Another benefit of using publicly available data sources is the possibility of linking the data with rankings based on journalistic or forensic estimates of wealth (see, e.g., Ring, Seim, and Zucman 2026 and Baselgia and Martínez 2026). I therefore supplement my data with wealth estimates from the magazine *Kapital*. Kapital has assessed the wealth of the wealthiest Norwegians for more than two decades and reports that roughly two years worth of labor goes into their annual forensic analyses (Kapital, 2023).

I verify that my data provides broad coverage of the top-400 wealthiest individuals per Kapital. In 2020, 388 of the top-400 individuals appear either in the shareholder register or in the municipal or zip-code level tax lists. The remaining individuals are typically Norwegian citizens who are foreign residents during the entire sample period.

Additional data sources. I also use data from *PitchBook*, scraped LinkedIn profiles from *Revelio Labs*, and patent application data from *Orbis*. For some analyses, I use data on startup grants from *Innovation Norway*. Similar to Innovate UK, Bpifrance, and the NSF's SBIR program, Innovation Norway is a government agency that provides competitive grants to innovative startups. I use receipt of these grants as a way to isolate startups from other firms that wealthy business owners invest in.

2.3 Measuring the wealth tax burden

To the extent that business owners finance wealth tax payments by extracting dividends from their firms, it is reasonable to consider the effective wealth tax as that which accounts for dividend taxes paid on the dividends required to pay the wealth tax. Consider the following equation in which the left hand side is the amount of taxes that must be paid and the right hand side equals the amount of dividends that are required to pay these taxes. If we consider this amount of dividends as the adjusted wealth tax (denoted $\widetilde{wtax_{i,t}}$), we may write

$$wtax_{i,t} + \widetilde{wtax_{i,t}} \cdot \tau_t^{div} = \widetilde{wtax_{i,t}}. \tag{1}$$

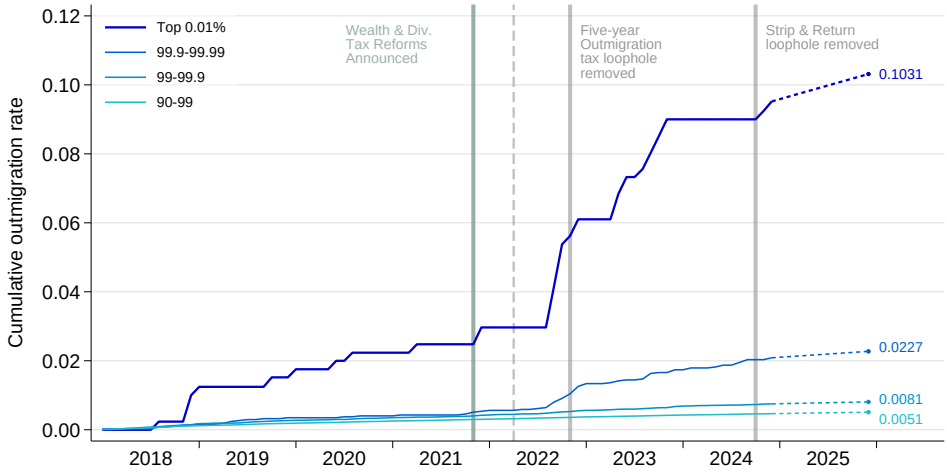
Rearranging, we get $\widetilde{wtax_{i,t}} = \frac{wtax_{i,t}}{1-\tau_t^{div}}$. To the extent that wealth primarily consists of unrealized capital gains, this dividend-tax inclusive measure is arguably the most natural measure to use when calculating, e.g., average tax rates on wealth. In any case, it is a useful measure for thinking about how the wealth tax affected the tax burden of entrepreneurs whose liquidity mostly resides inside corporate vehicles. Appendix A.5 discusses the role of partial dividend tax exemptions due to a rate-of-return deduction ("skjermingsfradrag").

3 Mobility responses

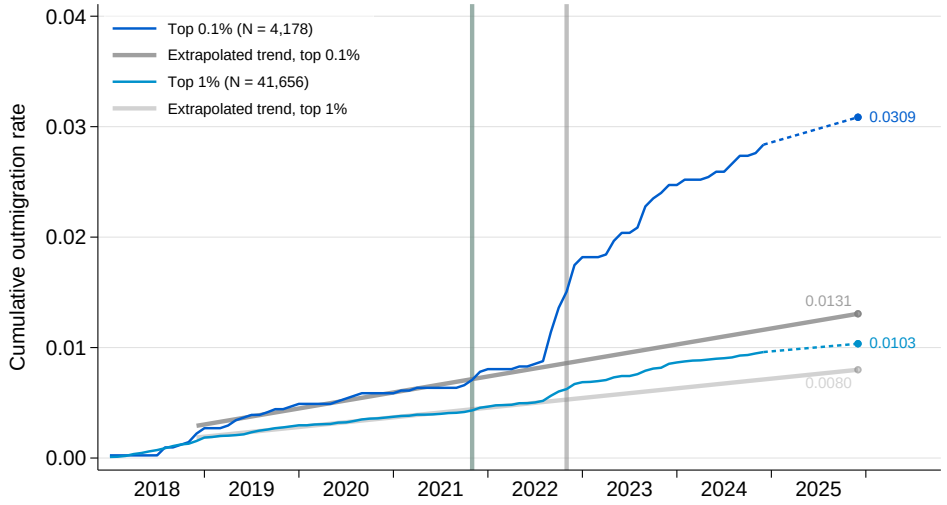
3.1 Migration responses across the wealth distribution, evidence from administrative data

FIGURE 2: OUTMIGRATION OF WEALTHY NORWEGIANS, ADMINISTRATIVE DATA

Panel A: Cumulative outmigration rates



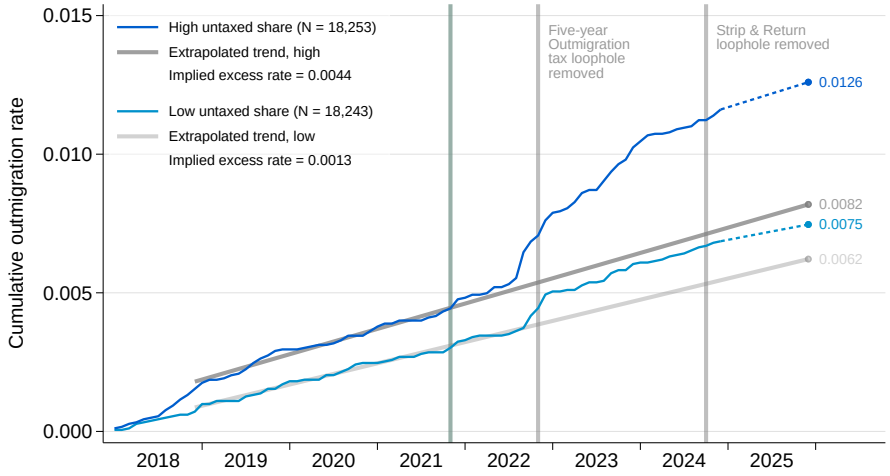
Panel B: Cumulative outmigration rates top 0.1% and 1%



Notes: This figure shows outmigration rates for different wealth groups, regardless of business ownership, around the 2021–2024 reforms. Panel (A) shows cumulative outmigration rates since 2017. Sample consists of those resident 01/01/2018, and wealth groups are assigned using 2017 tax return data. I impute exact migration dates from address registers. If someone outmigrates during 2020, and their address changes in May 2020, I assign May 2020 as the date of their move. Panel B pools the top groups in two ways: the top 0.1% and the top 1%. The dashed vertical is placed in April 2022 when the government confirmed that tighter exit tax rules were being developed. The dotted segments in 2025 indicate that only the year-end level is observed, not the timing of moves within the year. The underlying data comes from microdata.no. See Appendix Figure OA.2 for a version that omits the Top 0.01% for better visibility of the other wealth groups migration rates.

Figure 2 plots outmigration rates for individuals in different wealth bins. Panel A considers cumulative migration probabilities for those resident in Norway at the end of 2017. Using annual residency status and within-year address-change dates, I determine high-frequency (monthly) migration status. For the top 0.01% of the taxable wealth distribution, about 400 individuals, we see cumulative outmigration of 9.52% by the end of 2024. A large part of this can be explained by extrapolating pre-reform trends. Extrapolating the pre-reform trend implies that cumulative outmigration would have been about 4.5% in the absence of any reform, which suggests reform-induced outmigration of about 5%. For the [99.9%, 99.99%) group, we also see a sizable increase, albeit from much lower baseline outmigration rates. Appendix Figure OA.2 omits the top 0.01%, allowing us to better discern whether there is any effect below the top. While Panel A shows a clear effect for the 99.9–99.99% group, there is virtually no deviation from pre-reform trends for the 99–99.9% group. The 90–99% group remains exactly on trend.

Figure 3: Outmigration of Wealthy Norwegians in the Top 1%, Split By Potential Gains from Exploiting Outmigration Loopholes



Notes: This figure shows cumulative outmigration rates for two subgroups of the top 1% of the taxable wealth distribution. I divide the top 1% into two groups based on how much they would benefit from avoiding capital income taxation by exploiting migration loopholes. Specifically, I calculate a ratio of untaxed to total wealth in securities (incl. listed and unlisted stocks). My proxy for untaxed wealth is the undiscounted taxable value of securities minus the cumulative sum of dividend income and realized capital gains since 2006. The high group has an above-median share of untaxed wealth. The dashed lines indicates the lack of high-frequency data during 2025. The solid thick lines are extrapolated pre-trends (estimated using January 2019 through September 2021 data). Bin sizes may be slightly unequal due to privacy-induced noise on microdata.no (±10 individuals).

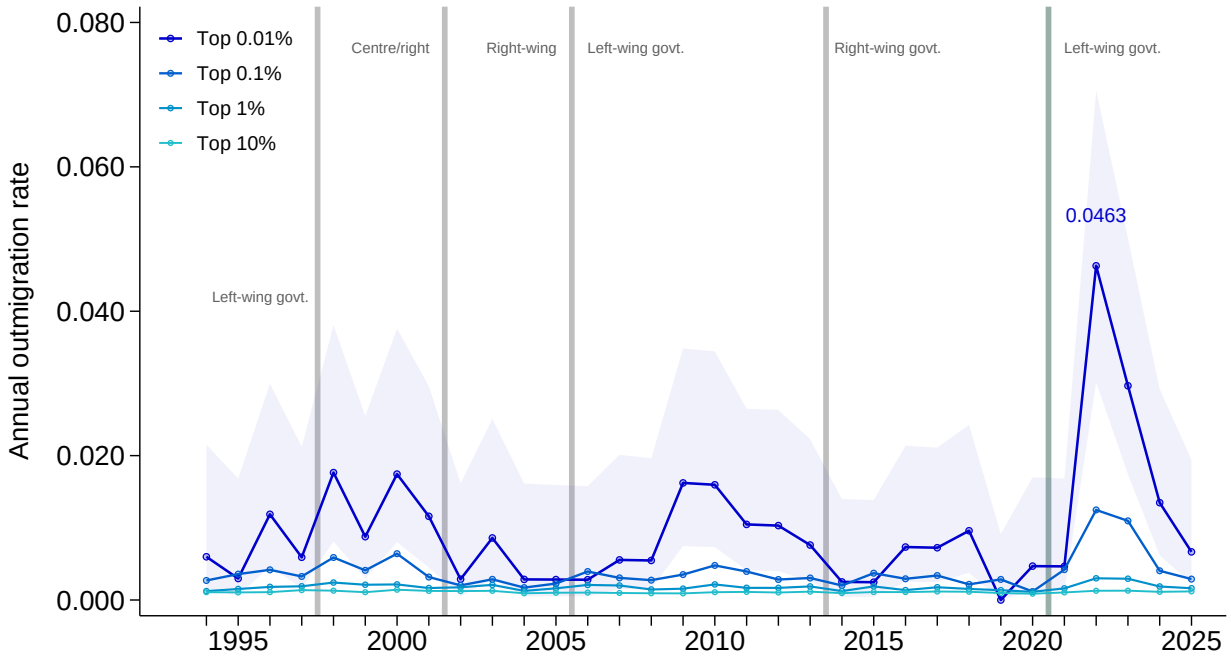
I next attempt to quantify the *number* of individuals who moved due to the reforms. Panel B of Figure 2 focuses on the inclusive top 0.1% and 1% groups. I include extrapolated pre-trends estimated using January 2019 to September 2021 data. Comparing the cumulative outmigration rates and the extrapolated counterfactuals indicate excess outmigration of 1.78% by the end of 2025 for the top 0.1% group, which in a sample of 4,178 equals about 74 individuals. The same

exercise for the top 1% group gives 0.24% or about 98 individuals.

Figure 3 considers cumulative outmigration rates for the top 1% of taxable wealth in 2017. I split this group into two based on a proxy for the share of their wealth that has not yet been subject to capital income taxation. The high-untaxed-wealth group has potentially more to gain from exploiting migration-related capital gains and dividend tax loopholes. Interestingly, I find very similar pre-trends during 2019–2021, but much stronger responses that persist until early 2024. The excess cumulative outmigration rate in the high-untaxed-wealth group is 0.44% (80 individuals), which is more than three times higher than the 0.13% rate in the low-untaxed-wealth group.

Figure 4 considers annual outmigration rates back to 1994. This demonstrates how the 2022–2023 outmigration rates at the top of the distribution are unprecedented in recent times.

Figure 4: Unprecedented Annual Outmigration Rates



Notes: This figure shows annual outmigration rates for a given year among those resident at the beginning of the year, and assigns wealth based on the prior year's tax data. The Top 0.01% rates come with 95% confidence intervals (95% Wilson score intervals). The statistics are calculated using the microdata.no platform.

3.2 Revenue effects in the administrative data

Panel A of Figure 5 shows total wealth tax revenues around the reform. The figure contrasts wealth taxes paid by stayers (bottom black line) with total counterfactual taxes if there were no outmigration since 2018. I compute the counterfactual tax amounts by assuming that outmigrants' tax revenues would have grown at the same rate as those with similar wealth

levels.

Panel B focuses on this wedge between actual and counterfactual tax payments. During 2022, the loss grows by about 0.5 BNOK, or about 0.4 BNOK net of the wealth taxes that emigrants continue to pay, consistent with the descriptive findings in Blandhol (2025).[11] By 2024, the estimated loss has grown to 0.86 BNOK. Netting out the predicted loss from a simple pre-reform linear extrapolation indicates that reform-driven outmigration reduced total wealth tax revenues by 0.65 BNOK. If we divide this by the estimated counterfactual tax payments of 31.97 BNOK from Panel A, we get a migration-related revenue loss of 2.05%. I can thus decompose the growth in wealth tax revenues from 2021 to 2024 in the following way

$$\% \text{ Change} = \underbrace{\left(1 + 0.7508\right)}_{\text{Counterfactual growth}} \quad \underbrace{\left(1 - 0.0205\right)}_{\text{Reform-driven outmigration}} \quad \underbrace{\left(1 - 0.0063\right)}_{\text{Other outmigration}} - 1 \approx 70.4\%, \tag{2}$$

where the counterfactual growth term assumes that the reform occurred but that no one outmigrated.[12] Reform-driven outmigration thus lowers the revenue increase from 75.1% to 71.5%, while the remaining reduction to 70.4% reflects outmigration predicted by pre-reform trends.

Panel B of Figure 5 also shows the migration-driven tax revenue losses net of taxes still paid by outmigrants. Some migrants moved to countries without a wealth tax or without a tax treaty that allows them to be immediately exempt from the wealth tax, and would thus still pay up to three years following outmigration. Accounting for this reduces the excess loss (loss minus extrapolated amount) modestly from 0.65 BNOK to 0.49 BNOK. However, this may be too conservative, since outmigrants' payments are presumably short-lived.

Using granular data on migration dates, I can show how the wedge changes at a high frequency. The modest trend between 2019 and 2021 is driven by modest and constant outmigration rates among the wealthy. The wedge then accelerates in the second-half of 2022, and before the five-year rule removal is effective (November 29th, 2022), the estimated loss has reached more than 0.50 BNOK. After the five-year rule removal, migration rates are more modest.

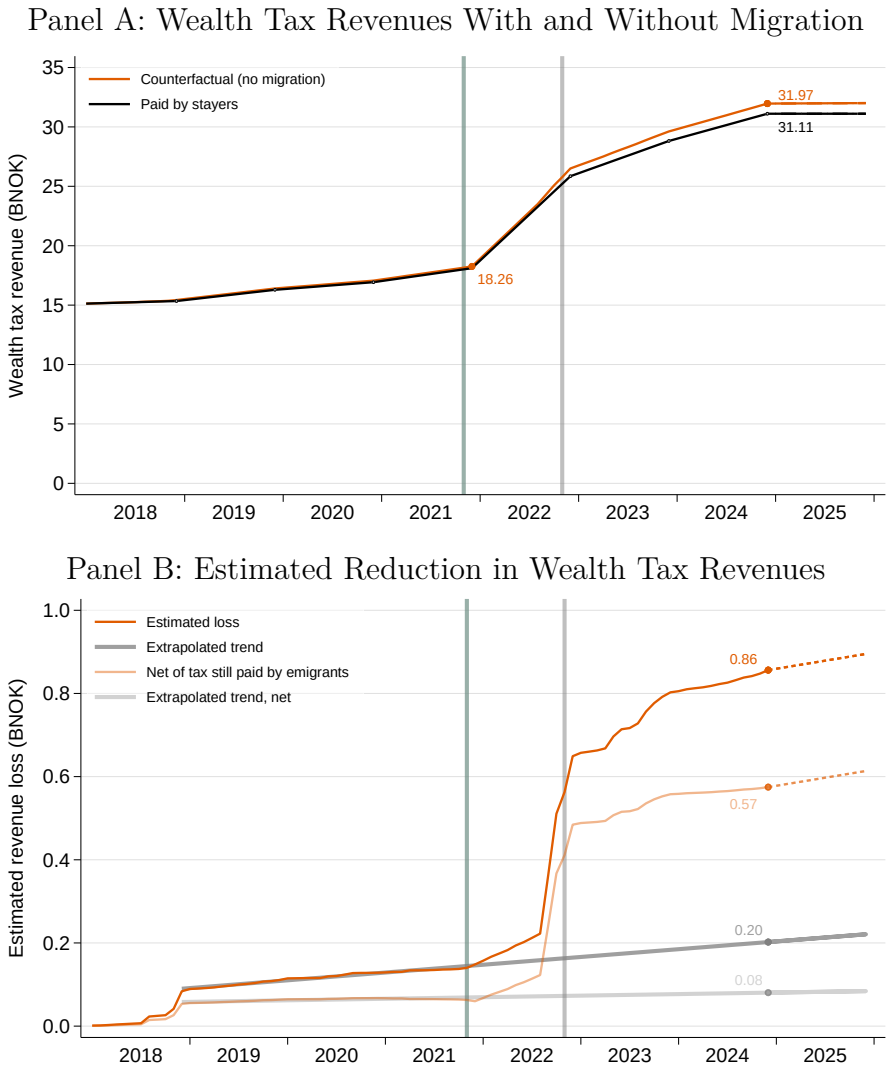
Appendix Figure OA.14 shows that there is no clearly discernible outmigration effect on income tax revenues. By the end of 2024, the difference between counterfactual (no outmigration) and actual revenues is about 1.7%. However, accounting for pre-period trends in outmigration reduces this difference to 0.12% of counterfactual income tax revenues (about 0.63 BNOK), and

[11] Blandhol (2025) documents that those who outmigrated during 2022 accounted for about 2.25% of 2021 wealth tax revenues. Dividing my estimated 0.4 BNOK net loss by the 18.26 BNOK base indicates a loss of 2.3%.

[12] The counterfactual growth term also reflects changes in wealth. As a comparison, applying the 2025 rather than the 2021 rules to the same 2025 population, which holds wealth fixed, raises wealth tax revenues by about 62% (Nygård et al., 2026).

by the end of 2025, the difference is zero.

FIGURE 5: TAX REVENUES AROUND THE REFORM, ADMINISTRATIVE DATA



Notes: This figure shows wealth tax revenues over time. Panel A, solid black line, shows aggregate wealth tax revenues paid by residents. The orange line adds the counterfactual payments of those who were resident on January 1st 2018 but subsequently emigrated. I assume that their year t tax payment equals their 2017 tax payment scaled up 2017-wealth-group-specific cumulative growth rate (calculated on stayers). I calculate annualized tax payments at the monthly level by interpolating between year ends. I assign exact migration dates using data on address moves during the year of migration. Panel B, top orange line, shows the estimated revenue loss, which is the difference between the counterfactual revenues and the revenues paid by residents. The dark gray line is an extrapolated trend. The second line (light orange) is the estimated loss when we include tax payments made by emigrants (who are often fully liable in the short term depending on the tax treaty with the destination country.) The circles and numerical values are the numbers as of year-end 2024. Post 2024 numbers are based on an assumed zero growth of tax payments, since I do not observe end-of-year 2025 tax payments nor within-year move dates for 2025. The underlying data comes from microdata.no.

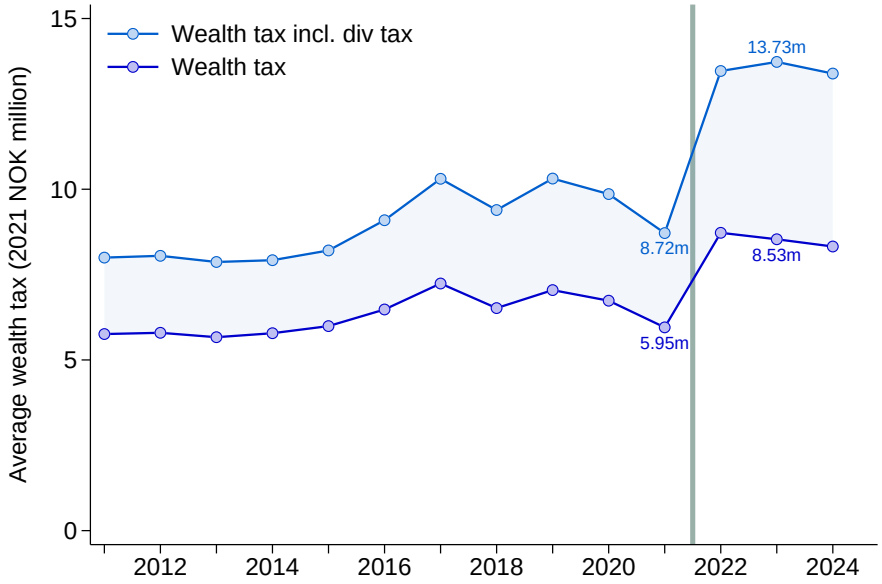
3.3 Changes in wealth tax exposure over time

Figure 6 plots two measures of wealth tax exposure for the top 0.01% over time. The figure shows the inflation adjusted amounts. We see that wealth tax payments were generally stable over time, hovering around 6 MNOK ($600,000), and increased almost 50% from 2021 to 2022.

From 2021 to 2023, the increase is 43%, or 57% when we account for dividend taxes on wealth tax amounts.

While the existing wealth tax literature considers elasticities with respect to wealth tax rates, it is not obvious that this is the right comparison. If potential outmigrants face (possibly non-pecuniary) costs of moving that do not scale with their wealth, absolute tax amounts may be the correct exposure measure. Appendix Figure OA.5 contrasts the two measures, and finds that effective wealth tax rates were in fact higher during the 2012–2013 period than they are after the 2021–2024 reforms. The perhaps more correct measure relates tax payments to foregone investment, in which case we should deflate tax amounts by an investment price series. Appendix Figure OA.6 shows that this yields quantitatively similar changes to wealth tax exposure. Exposure is mostly flat during 2011–2021 and increases by 39% from 2021 to 2023, or 53% when including dividend taxes.

Figure 6: Wealth Tax Exposure Over Time for the Top 0.01%



Notes: This figure shows inflation-adjusted wealth-tax amounts over time for those in the top 0.01% of taxable wealth in 2017. The underlying statistics are produced with microdata.no. See Appendix Figure OA.6 for similar results using an investment price deflator instead of a consumption price deflator.

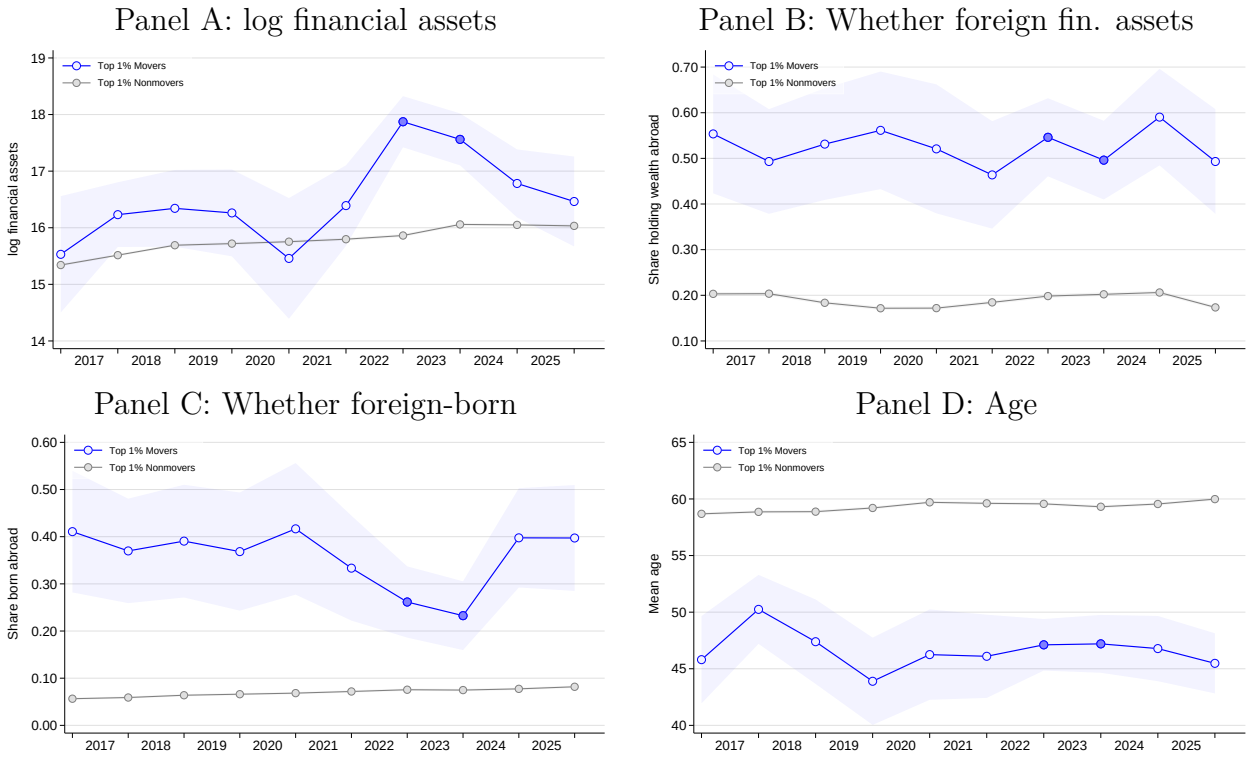
3.4 Outmigrant characteristics

Figure 7 considers whether those who outmigrated during the exodus period of 2022–2023 differed from other outmigrants or from stayers. Panel A considers log financial assets. While movers generally have comparable amounts of financial assets as stayers during 2016–2021, the 2022–2023 outmigrants were considerably wealthier. These outmigrants have about 1.5–2 log points more financial wealth than stayers.

Panel B shows that 2022–2023 outmigrants do not appear to have a stronger foreign connection than other outmigrants in terms of direct ownership of foreign financial assets. Panel C shows that 2022–2023 outmigrants were more likely than stayers to have been born abroad, which is consistent with prior studies showing lower migration elasticities of natives (Akcigit, Baslandze, and Stantcheva 2016, Kleven, Landais, Muñoz, and Stantcheva 2020). Interestingly, however, the 2022–2023 outmigrant cohorts were less likely to be foreign-born than other migration cohorts.

Panel D shows that outmigrants are generally younger (about 45) than stayers (about 60), and that 2022–2023 outmigrants are not special. This contrasts with Moretti and Wilson (2023) who document higher migration elasticities among older individuals, presumably because that setting revolves around end-of-life estate taxation rather than recurring taxation in my setting.

FIGURE 7: OUTMIGRANT CHARACTERISTICS IN THE ADMINISTRATIVE DATA



Notes: This figure shows characteristics measured at $t-1$ for those who moved during year t. The lightly-shaded gray lines provide the characteristics of nonmovers. The sample consists of those in the top 1% in the prior year (top 0.1% or narrower hits privacy constraints due to few movers).

An additional interesting observation is that by 2024, movers again appear to resemble pre-reform movers on observables such as log financial wealth and whether foreign born. This corroborates the other findings that the special response to the tax reforms was essentially over by 2024 when most of the migration-related tax loopholes were removed (but the wealth tax

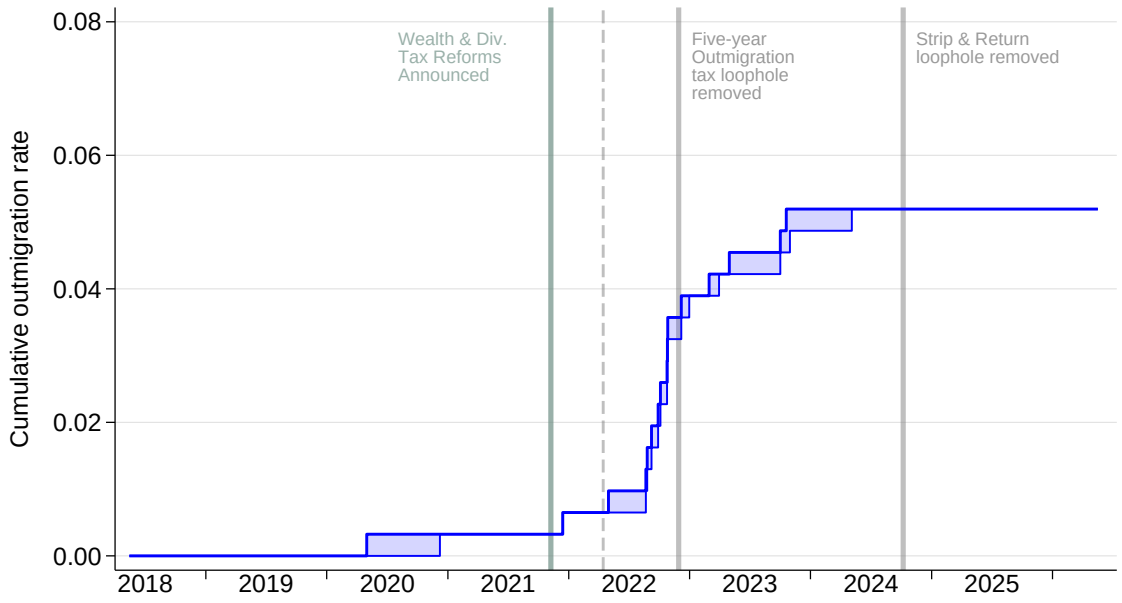
was still in place).

3.5 Outmigration of High-Profile Individuals

The outmigration episode following the 2021–2024 reforms saw widespread media attention. In an attempt to distinguish between wealth-tax and exit-tax motivated outmigration, I use publicly available data on migration events of high-profile individuals to study when these individuals moved. Figure 8 considers persons in *Kapital*'s list of Norway's 400 wealthiest individuals. This reveals that most of the spike in outmigration is concentrated in the months directly before the five-year loophole was removed, indicating that exit taxation was an important cause for the rapid outmigration event. Nevertheless, Appendix Figure OA.3 considers individuals in the business newspaper *Finansavisen*'s list of high-profile "bellwether" investors, and finds a more sluggish response. These investors primarily outmigrated after the exit-tax loophole was removed.

Figure 8: Outmigration of High-Profile Billionaires

Panel A: Rich-List Billionaires



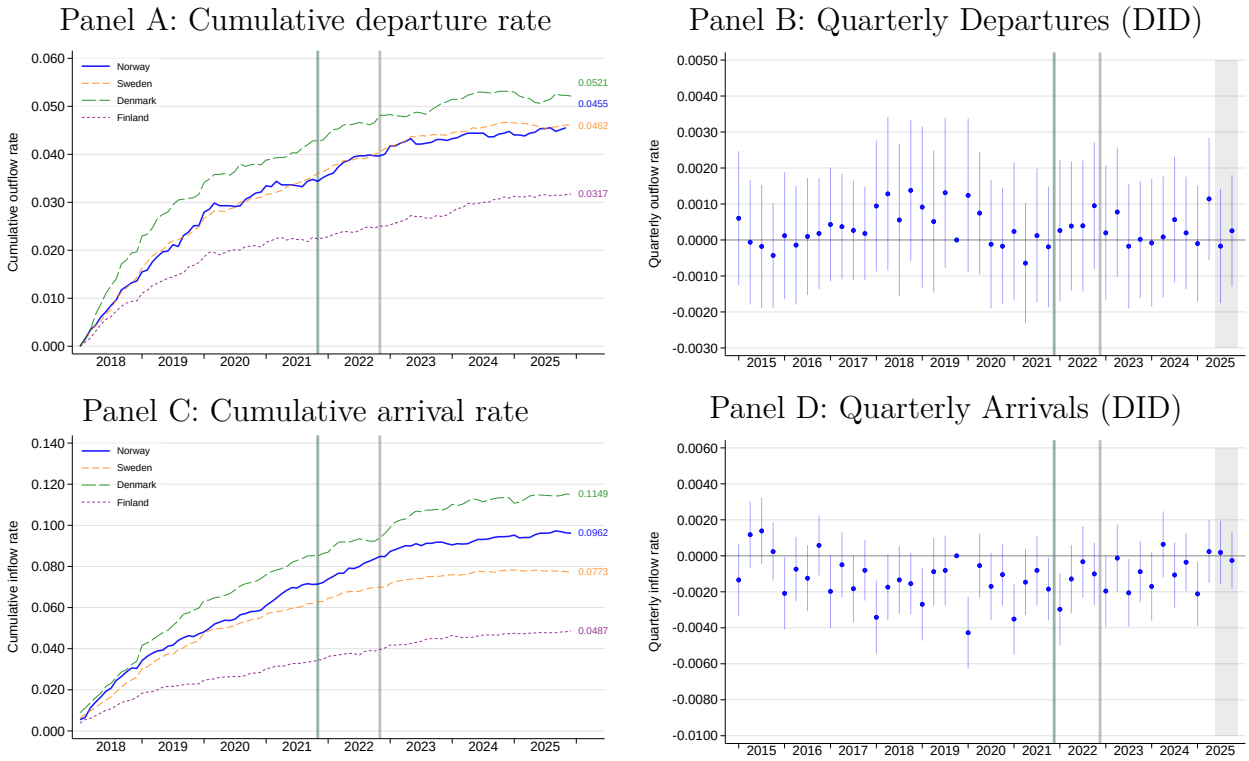
Notes: This figure shows cumulative outmigration rates for high-profile billionaires: those who were resident in Norway end-of-year 2017 and who appeared in the Kapital 400 wealth ranking during 2012–2017. 2017 Residence is determined using the shareholder register. Migration dates are determined using the shareholder register, LinkedIn data, and newspaper articles. The dashed vertical is placed in April 2022 when the government confirmed that tighter exit tax rules were being developed. The shaded area indicates ambiguous migration dates (e.g., a range of dates from newspaper articles).

3.6 Mobility of Startup Entrepreneurs and Inventors

I also assess whether startup entrepreneurs and inventors, the potential drivers of innovation and growth, left the country in response to the reforms. Even if owners of young firms

are not directly affected through tax liabilities (Thoresen et al., 2022), entrepreneurial or innovative activity may follow wealthy financiers abroad. For these analyses, I rely on scraped LinkedIn profiles from *Revelio Labs*, which allows me to contrast the mobility of entrepreneurs and inventors in Norway with those in the other Nordic countries.

FIGURE 9: MIGRATION OF STARTUP ENTREPRENEURS



Notes: This figure shows migration rates of startup entrepreneurs across the Nordic countries. Panel A plots the share of founders residing in a given country as of 2017 that subsequently move to another country. Residence and mobility is determined using the location of the most-recent employer on a founder's LinkedIn profile at any point in time. Panel C plots *inflows*: the number of arriving founders, divided by the stock of founders in the destination country as of 2017. These analyses rely on LinkedIn data from Revelio Labs. An individual is classified as an entrepreneur using LinkedIn occupation codes, and is limited to those classified as startup or tech founders, founders at tech firms, serial founders, founders who list entrepreneurship skills, founders at *PitchBook* deal-active firms, or founders at patenting firms. Panels B and D plot Norway-by-quarter difference-in-differences (DID) coefficients from a linear probability model with quarter fixed effects, using Sweden, Denmark, and Finland as the control group. The reference quarter is 2019Q4, standard errors are clustered by person, and the shaded quarters are affected by incomplete updating of LinkedIn profiles.

Panel A of Figure 9 considers startup entrepreneurs identified via their LinkedIn profiles. I consider their move date to be the first date of a new job outside their original country. I contrast the outmigration rates of Norwegian, Swedish, Danish, and Finnish entrepreneurs. Interestingly, Norwegian and Swedish entrepreneurs outmigrated at nearly identical rates between 2018 and 2025, with no observable differences after the reforms. Panel C considers the entrepreneurial *arrival* rate, defined as the number of foreign entrepreneurs moving to Norway divided by the ex-ante number of Norwegian-based entrepreneurs. I find no indication that entrepreneurial inflows were reduced relative to other Nordic countries.

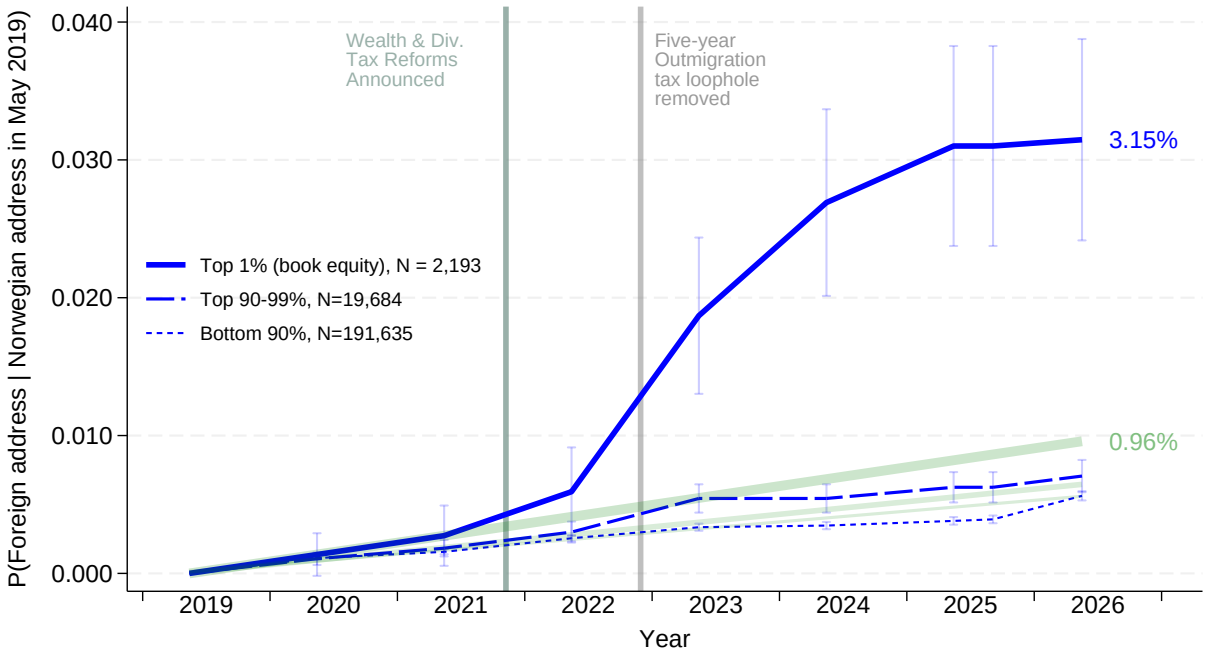
Appendix Figure OA.4 merges patenting data with LinkedIn profiles to similarly study the

migration rates of inventors. This similarly reveals no indication of excess outflows or reduced inflows.

3.7 Outmigration of business owners

Effect on business owner outmigration. The public shareholder register contains information on the most recent country of residence for all shareholders. I supplement the data with historical versions of the public shareholder register to obtain a panel of residency starting May 2019.[13]

FIGURE 10: BUSINESS OWNER OUTMIGRATION IN THE SHAREHOLDER REGISTER



Notes: The figure shows the share residing abroad among individuals resident in Norway around May 2019. Individuals who appear as foreign residents and subsequently leave the data remain coded as foreign residents. Observed returns to Norway are counted as domestic residence. The green lines extrapolate linear trends fitted to the 2019–2021 observations. Individuals are grouped by their average directly owned book equity during 2016–2018. See Appendix Figure OA.9 for a version that considers the top 5% of the distribution.

Figure 10 limits the sample to those residing in Norway around May 2019. The figure shows the share with a foreign address in subsequent years. Individuals who appear with a foreign address and subsequently leave the register are coded as still being foreign residents. Observed returns to Norway are counted as domestic residence. I split the sample into three bins based on their average directly owned book equity in private firms during 2016–2018. We

[13]Each vintage of the public shareholder register typically contains the addresses/residency of a shareholder as of the date when the register is updated with new data. The 2018 shareholder data was made available in the spring of 2019, thus the 2018 vintage contains spring 2019 addresses. To get a panel, I combine several vintages (differently-dated downloads) of the shareholder register.

see that those in the top-1% (of book equity) leave Norway at higher rates even before the 2021–2024 reforms. This group consists of 2,193 individuals. The green line extrapolates the 2019–2021 linear trend to give a first-order approximation of the counterfactual outmigration rates absent the reforms.

We see that the share residing abroad is about 3.15%, which is 2.19 percentage points higher than the extrapolated trend. In terms of the number of individuals, there were about 48 excess outmigrants in the top-1% group by May 2026. In the other groups (top 90–99% and the bottom 90% of book equity), there is no indication of any excess outmigration. Appendix Figure OA.9 expands the top bin from the Top 1% to the Top 5% to assess whether we are missing potential outmigrants in the 95th to 99th percentile of business owners. However, this expansion only increases the estimated number of outmigrants to 56 (as opposed to 48) individuals.

Economic footprint. Table 1 provides statistics on the firms owned by outmigrants. I consider three groups of individuals. Outmigrants in the top 1%, all business owners in the top 1%, and all Norwegian business owners. Panel A allocates firm-level measures of economic activity, including employment, wage costs, investment, and revenue, to individuals according to their ownership share. Panel B considers firm-level statistics for firms owned $> 20\%$ by the individuals considered in panel A.

Some key statistics quickly emerge by considering outmigrants' share of total firm-level values. Contrasting top-1% outmigrants to all business owners in the top 1% reveals that outmigrants account for 5.45% of allocated employment, 5.54% of wage costs, 7.49% of investment, and 4.46% of revenue (Panel A). These shares are all considerably higher than 2.19%, which is their share of individuals. If we contrast the firms of top-1% outmigrants instead to all Norwegian-owned firms (Panel B), we find that they account for a sizable share of economic activity: 1.24% of employment, 1.41% of wage costs, 10.64% of investment, and 2.53% of revenue. This investment share is largely driven by one very large firm. Removing the largest and lowest observations yields a share of 1.88% (2.53% in Panel A).

Economic activity in outmigrants' firms. Jakobsen, Kleven, Kolsrud, Landais, and Muñoz (2024) document substantial effects of outmigration on the owners' firms. Value added, turnover, and employment all decrease by about 20–35% within 3–4 years of the migration event. This effect is primarily driven by firm exits. Within three years of outmigration, a previously-held firm is about 25 percentage points less likely to be active. These effects are noteworthy. I therefore study what happens to outmigrants' firms in Norway as well. In the Norwegian setting, these effects can be more directly identified since the reform increased outmigration.

TABLE 1: OUTMIGRANTS' ECONOMIC FOOTPRINT

Panel A: Amounts allocated to individuals by ultimate ownership shares						
Monetary amounts in MNOK	May 2022–May 2024 Outmigrants		All Top 1%		All Norw. owners	
	Total	Average	Total	Average	Total	Average
Employees	8,644	180.1	158,584	73.2	1,145,330	3.9
Wage bill	5,060	105.4	91,418	42.2	494,857	1.6
Tangible investment	3,838	80.0	51,238	23.7	128,941	0.4
Tangible investment*	1,229	25.6	48,630	22.5	127,842	0.4
Operating revenue	32,788	683.1	735,340	339.3	2,537,787	8.3
Individuals	48		2,193		313,016	
Panel B: Firms with at least one individual owning more than 20%						
	May 2022–May 2024 Outmigrants		All Top 1%		All Norw. owners	
	Total	Average	Total	Average	Total	Average
Employees	13,357	10.9	184,908	10.5	1,076,504	3.8
Wage bill	6,332	4.9	102,409	5.5	447,606	1.5
Tangible investment	12,724	10.6	60,268	3.4	119,620	0.4
Tangible investment*	2,067	1.7	49,612	2.8	109,687	0.4
Operating revenue	59,667	46.2	875,920	46.9	2,361,268	7.9
Firms	1,308		18,898		306,126	

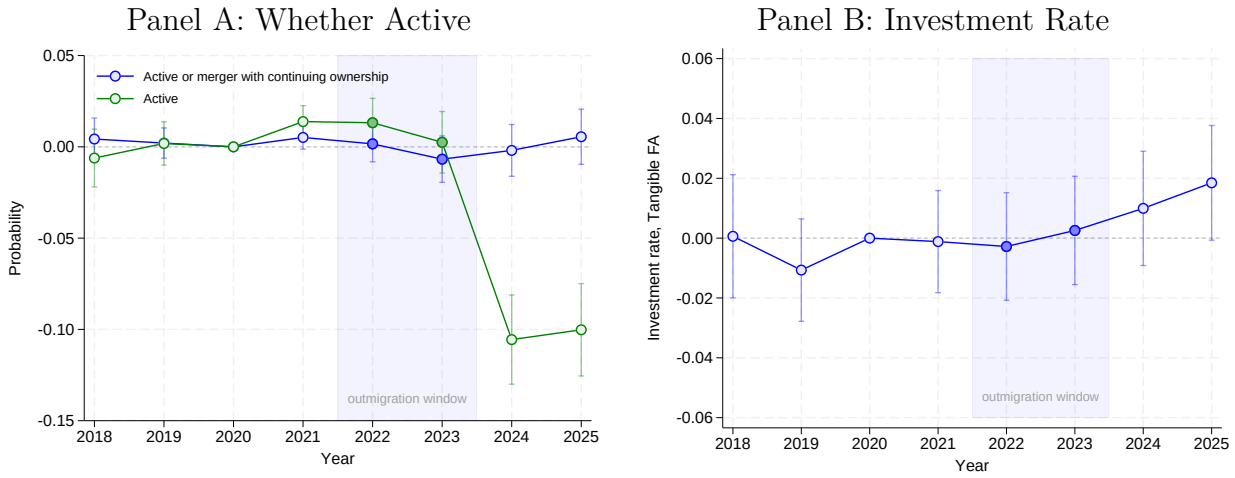
Notes: This table provides statistics on the economic activity of firms owned by outmigrants. I consider outmigrants in the top 1% (sorted by book equity) during the main May 2022–May 2024 wave. All amounts are measured in 2020. *Excludes the firms with the lowest and highest 2020 investment in each column.

Figure 11 shows results from a difference-in-differences analysis, comparing firms whose owners outmigrated during the exodus of May 2022–May 2024 with stayers. Similar to Jakobsen et al. (2024), I consider all firms owned $> 20\%$ by an individual who either outmigrated or stayed. The base sample is firms owned by individuals in the top 1% of the business owners (sorted by book equity). Panel A considers whether a firm remains active in business registers. This is the outcome that explains most of the effects in Jakobsen et al. (2024). The initial estimates are large. An outmigrant-owned firm is more than 10 percentage points less likely to be active immediately after the 2022–2024 outmigration window. This result, however, is entirely driven by merger-related activity rather than pure liquidation or bankruptcy. The Orbis historical data enriches the statuses available in the business registries by appending information from creditor announcements. These creditor announcements (published via the business registry office) detail the type of dissolution, distinguishing between, e.g., liquidation, mergers, and acquisitions. I supplement this data with the full actual announcements from which I extract the identifiers of the successor firm (i.e., the merged corporate entity). These supplementary data allow me to identify and disregard firms that become notionally inactive due to being merged into another entity with ownership continuity. This refinement uncovers

that there is no negative effect on whether the firm is still active.

Interestingly, Panel B shows no sign of reduced investment for firms that remain active. This is either consistent with no adverse effects of reform-driven outmigration or with the notion that outmigration allows owners to avoid wealth taxes. Avoiding wealth taxation may allow outmigrants' firms to be less adversely affected by the wealth tax. However, this presumes that the wealth tax reduced investment in wealth-tax-exposed firms, which I examine in Section 4.6.

FIGURE 11: ECONOMIC ACTIVITY OF OUTMIGRANTS' FIRMS



Notes: The figure shows differential characteristics of outmigrant-owned versus non-outmigrant owned firms over time. The sample contains firms owned > 20% in 2018 by individuals in the top 1% of business owners (sorted by book equity). I consider the firms owned by those who outmigrated during May 2022 to May 2024. Comparison firms are those whose top-1% owners remain in Norway through May 2026. Regressions include a treatment indicator, year fixed effects, and treatment-by-year interactions, with 2020 as the reference year. The shaded blue area roughly corresponds to the timing of outmigration. In Panel (A), we consider whether the firm is still active. The bottom green line considers the probability of the firm still being active, counting firms that exited due to M&A as inactive. The top blue line includes active as well as those dissolved due to M&A activity in which there is ownership continuity, defined as: (i) prior-year direct owners who each owned > 20% jointly own > 50% of the successor, (ii) prior-year ultimate owners who each owned > 20% jointly own > 20% of the successor, or (iii) the successor owned > 20% of the firm in the prior year. Standard errors are clustered at the firm level. Panel B considers the effect on the investment rate in tangible fixed assets (TFA), defined as in Section 4.6.

Note that we need additional assumptions to infer that the reforms did not induce exits or reduce investment rates. Exits of compliers (those who left due to the reform) may have different effects than exits of always-takers (those who would have left regardless). Hence, the necessary assumption is that always-taker effects are not sufficiently positive to offset the negative effects from compliers. The findings in Jakobsen et al. (2024) support this assumption.[14]

[14]Testing this assumption is challenging in my setting. There is very little outmigration in the years before the reform and data on shareholder residency is only available from 2018, implying that there are few outmigrants from which to estimate the effect from always-taker exits.

4 The Effect on Businesses

4.1 The User Cost of Capital

This section provides a formal mapping between the parameters that determine personal taxes (e.g., dividend tax rate, wealth tax schedule) and the user cost of capital from the perspective of the firm. The goal is not a comprehensive treatment of how personal taxes affect corporate investment, but rather a stylized mapping that highlights the theoretically ambiguous effect that personal taxation has on firm investment. The analysis builds on the user-cost framework of Jorgenson (1963) and Hall and Jorgenson (1967) and derives the required return from the owner's tax treatment and intertemporal problem. Since the owner cannot borrow against future business income, investment and consumption decisions are not separable (Hirshleifer, 1958). Hence, the EIS matters for the user cost.

Consider an established owner with \$1 of retained earnings. At date 0, the owner chooses $k \in (0, 1)$ to retain in the firm and distributes the remaining $1 - k$ as dividends. This is the retained-earnings margin. The firm earns annual operating income $F(k)$ before depreciation and corporate income tax for T years and is liquidated at the end of year T. Capital depreciates at rate δ, and corporate income is taxed at rate τ^{CIT}. The firm replaces depreciation but cannot otherwise adjust its capital stock or hold financial assets. It distributes its after-corporate-tax profit each year. The owner pays dividend tax τ^{Div} on these distributions and the liquidation proceeds. The owner also pays annual wealth tax $\theta^S \tau^w k$ on the shares, where θ^S is their marginal wealth-tax assessment fraction.

The owner may save personally but cannot borrow. Let $s_{t+1} \geq 0$ denote the amount saved at date t. Personal saving earns r^{save} after personal income tax and has wealth-tax assessment fraction θ^P. The owner consumes the liquidation proceeds over J retirement years, beginning at date T. Let c_t denote consumption at date t, let $\beta \in (0, 1)$ denote the discount factor, and let u be increasing, differentiable, and strictly concave. I compare two tax systems in which all tax parameters remain constant over the owner's remaining life and solve the owner's consumption and saving choices under each.

Let R be the annual marginal revenue product required for the owner to retain k in the firm. Appendix A shows that

$$R = \delta + \frac{1}{1 - \tau^{CIT}} \left[\frac{\theta^S \tau^w}{1 - \tau^{Div}} + \frac{u'(c_0) - \beta^T u'(c_T)}{\sum_{t=1}^{T} \beta^t u'(c_t)} \right]. \tag{3}$$

The direct tax term is consistent with the new view of dividend taxation (King 1974, Auerbach 1979, Bradford 1981, Auerbach and Hassett 2003). When retained earnings are the marginal

source of finance, dividend tax applies to both the foregone date-0 distribution and subsequent payouts. It appears directly only in the wealth-tax gross-up. At the pre-reform optimum, $F'(k) = R$. To isolate the reform-induced change in the owner's required return, I evaluate equation (3) under the post-reform tax system while holding k at its pre-reform value.

When the owner saves personally at every business date, the Euler equations reduce equation (3) to

$$R = \delta + \frac{r^{save} - \theta^P \tau^w + \dfrac{\theta^S \tau^w}{1 - \tau^{Div}}}{1 - \tau^{CIT}}. \tag{4}$$

If $s_{t+1} = 0$ at date t, the Euler condition is an inequality, and the required return also depends on the consumption profile in equation (3).

Suppose that the owner holds no personal saving prior to retirement ($t \leq T$) and that $u'(c) = c^{-1/\sigma}$, where σ is the elasticity of intertemporal substitution (EIS). Differentiating with respect to $\theta^S \tau^w$, while holding k, τ^w, and τ^{Div} fixed, gives

$$\left.\frac{\partial R}{\partial(\theta^S \tau^w)}\right|_k = \frac{\sigma - \sigma^*_{T,J}}{\sigma(1 - \tau^{CIT})(1 - \tau^{Div})}, \tag{5}$$

where $\sigma^*_{T,J}$ is an EIS cutoff defined in Appendix A.3. The cutoff is a marginal-utility-weighted average of the elasticities of annual business cash flow and liquidation proceeds with respect to k. If both are positive and increasing in k, $F(0) = 0$, and F is strictly concave, then $0 < \sigma^*_{T,J} < 1$. A higher wealth-tax assessment fraction for shares can therefore lower, leave unchanged, or raise the required marginal revenue product, depending on the EIS.

The reform's effect on the user cost of capital in a calibrated model. I specify annual operating income as $F(k) = Ak^\gamma$. Chodorow-Reich (2025) reports a capital revenue elasticity of 0.22–0.35 and a labor revenue elasticity of 0.65. I use the midpoint of the capital range, which gives $\gamma = 0.285/(1 - 0.65) = 0.8143$. Table 2 reports the calibration parameters in panel A and the implied user-cost changes in panel B. Under the full reform and resulting saving paths, the numerical change crosses zero at an EIS of approximately 0.661.[15] The final column uses -0.5, the midpoint of the preferred range in Chodorow-Reich (2025) for the short-run semi-elasticity $d(I_t/K_{t-1})/d\log R_t$ in the Coen (1969) specification.

I consider several values of the EIS. Jakobsen, Jakobsen, Kleven, and Zucman (2020) calibrate a life-cycle model to responses to a reform in Denmark in the 1980s, and find implied values of the EIS ranging from 2 to 6. Hence, these are included in the table below. Estimates from Norway lie on both sides of the 0.66 threshold. Responses to wealth taxation imply an EIS

[15] With partial assessment of personal saving ($\theta^P = 0.5$), the threshold rises to about 0.76 (Appendix Table OA.2).

of 0.05–0.20 (Ring 2025; Ring and Thoresen 2025), whereas Holm, Jamilov, Jasinski, and Nenov (2026) estimate an EIS of 1.5 for firm owners based on responses to an announced dividend tax increase.

Table 2: 2021–2024 Reforms and the User Cost of Capital

Panel A: Calibration and tax parameters

Parameter	Value	Parameter	Value
Horizon	$T = 20,\ J = 20$	Discount factor	$\beta = 0.975$
Personal saving return	$r^{save} = 0.03$	Saving assessment fraction	$\theta^P = 1$
Required-return premium	$\pi = 0.02$	Personal borrowing	Not allowed
Depreciation	$\delta = 0.10$	Corporate income tax	$\tau^{CIT} = 0.22$
Share assessment fraction	$\theta^S : 0.55 \to 0.80$	Wealth tax	$\tau^w : 0.0085 \to 0.0110$
Dividend tax	$\tau^{Div} : 0.3168 \to 0.3784$	Investment semi-elasticity	-0.5

Panel B: Implied change in the user cost of capital

EIS	Baseline k	R^{post}	ΔR (pp.)	$100 \times \Delta \log R$	$\Delta(I/K)$ (pp.)
0.20	0.9387	0.1511	−1.090	−6.967	3.483
0.25	0.9393	0.1518	−1.017	−6.487	3.244
0.50	0.9422	0.1587	−0.331	−2.065	1.032
1.00	0.9476	0.1655	0.354	2.163	−1.081
2.00	0.9568	0.1690	0.698	4.220	−2.110
6.00	0.9804	0.1712	0.918	5.515	−2.757
		Assuming the owner saves at every date, $s_{t+1} > 0$			
–	–	0.1425	0.617	4.428	−2.214

Notes: Panel A reports the calibration parameters. All rates are annual. Arrows distinguish pre- and post-reform values. The personal saving return is after personal income tax and before wealth tax. $\theta^P = 1$ indicates that personal savings is not subject to an assessment discount. The owner operates the firm for $T = 20$ years and lives for an additional $J = 20$ retirement years. Panel B solves the owner's problem under two tax schedules: pre-reform and post-reform. I set the baseline user cost to $R^{pre} = 0.16198$ by imposing a two-percentage-point premium over the return on private savings. For each EIS, I choose A such that equation (3) gives $F'(k) = R^{pre} = 0.16198$. I then hold A and k fixed, impose the reform, and reoptimize the consumption and saving path. The final row assumes that the owner saves at all t under both tax schedules, and thus R is pinned down according to equation (4), which implies $R^{pre} = 0.1363$. The final column reports the implied percentage-point change in I/K, calculated as $-0.5 \times (100\Delta \log R)$ using the midpoint of the short-run semi-elasticities reported by Chodorow-Reich (2025). Appendix A presents the model, derivations, and optimal saving paths.

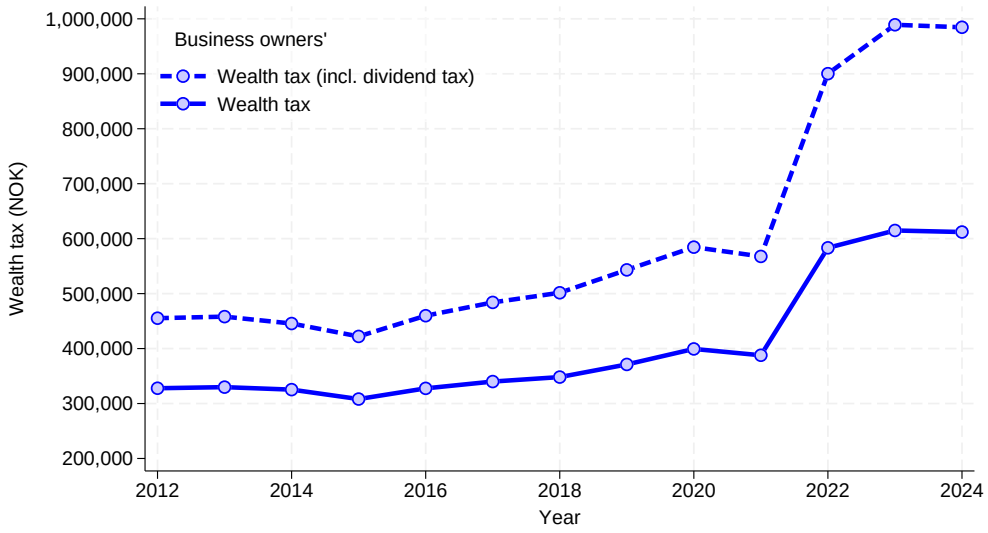
4.2 Business owners' wealth tax exposure

The changes to the tax code and broad media coverage indicate that business owners' wealth taxes went up significantly due to the reform. I illustrate the increased wealth tax burden in Figure 12. The figure considers business owners observed in the shareholder register. The main sample restriction is that individuals appear in the top-100 municipal tax lists when sorted by wealth. This restriction thus selects on paying non-trivial amounts of wealth taxes.

The increase around the reform is nevertheless interesting because it tells us how much more those who paid the most wealth tax paid because of the new rules. I find that the average amount increased from about NOK 390,000 ($39,000) to NOK 610,000 from 2021 to 2024, or about 60%.

Figure 12 also considers the dividend-tax inclusive wealth tax measure. This assumes that wealth taxes are paid for with dividends that trigger dividend taxation (see equation 1). This amplifies the effect of the reform, partly due to a pure scaling effect, and partly due to increased dividend tax rates. Wealth tax exposure increases by about 73%, from NOK 570,000 ($57,000) in 2021 to about NOK 980,000 in 2024, once we account for dividend taxation. Appendix Figure OA.7 shows total tax less calculated wealth tax and implied dividend tax. This remainder (personal income tax excl. wealth-tax-implied dividend tax) does not show a comparable increase during 2022–2024.[16]

Figure 12: Higher Wealth Taxes for Business Owners



Notes: This figure provides average wealth taxes calculated from reported taxable wealth and each year's tax schedule for individuals in the municipal top-100 lists (available 2012–2024). Individuals are designated business owners if they own at least 50% of an LLC (using data from the shareholder register). The dashed line provides wealth tax amounts inclusive of implied dividend taxes. If wealth taxes are financed with dividends, the dividends must cover both wealth tax and the resulting dividend tax (see equation 1).

4.3 Empirical specification

I infer the effects of the 2021–2024 reforms by comparing Norwegian-owned firms with other firms. The main setting (*Setting 1*) compares Norwegian-domiciled firms by whether owners are subject to the wealth tax. The treatment group consists of firms for whom I

[16]But it does show a spike in 2021 due to anticipatory responses to the increase in the dividend tax, which led to higher dividend payments and thus income tax payments in 2021.

assign at least 67% of iterated (ultimate) ownership to Norwegian individuals. The control group consists of firms for whom I can assign less than 33% ownership to individuals resident in Norway or domestic partnerships, and where (i) no individual Norwegian owns more than 10%, (ii) Norwegian or local governments own at most 50%, (iii) the firm appears in the 2018 shareholder register. The second setting compares non-Norwegian-domiciled firms owned by Norwegian persons with non-Norwegian firms owned by other foreign owners (e.g., I compare Norwegian-owned and Swedish-owned German firms). This classification makes use of the historical Orbis ownership data on global ultimate owners. *Setting 3* considers Nordic-domiciled firms, contrasting $\geq$ 67%-Norwegian-owned Norwegian firms with firms in the other Nordic countries (for whom I cannot assign Norwegian ownership). Treatment status is defined using 2018 ownership, so the estimates capture effects on firms whose owners were exposed to the reform, including owners who later emigrated. Appendix B.4 provides additional detail on sample restrictions.

The results are based on estimating the following regression equation on firm-year (f, t) level data.

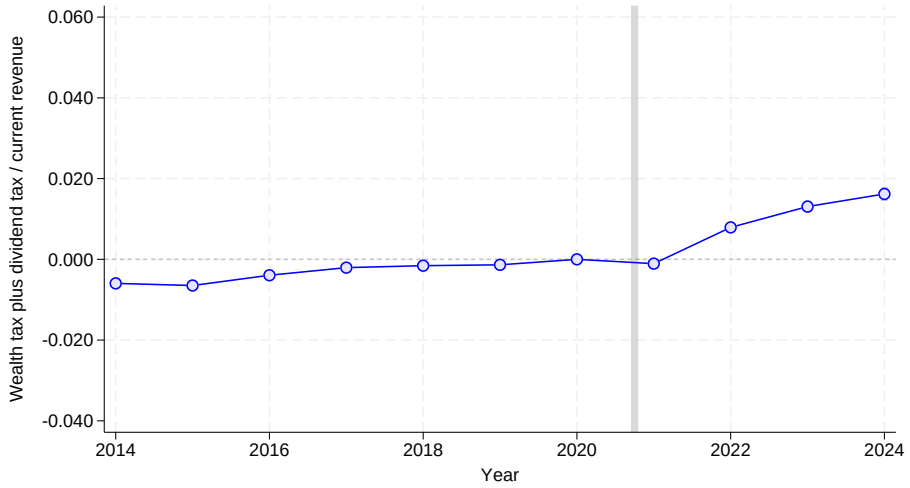
$$y_{f,t} = \beta_t Treat_{f,2018} + \alpha_f + \gamma_{M,t} + \varepsilon_{f,t}, \tag{6}$$

where α_f are firm fixed effects, $\gamma_{M,t}$ removes time-varying fixed effects, and 2020 is the omitted year. For setting 1, I take out 3-digit NACE code by year fixed effects, 2013–2014 revenue decile by year fixed effects, 2014 tangible-fixed asset group by year fixed effects, and 2014 age-category by year fixed effects. Settings 2 and 3 use similar fixed effects, with adjustments for sample sizes and data availability. Appendix B.4 provides details. Outcome construction is described in Appendix B.5.

4.4 Effect on firm-level wealth tax exposure

While the descriptive analyses demonstrate a large increase in wealth tax payments for entrepreneurs, it is useful to quantify this effect using the same empirical methodology that I use to consider real effects on firm outcomes.

Figure 13: Effect on Wealth Tax Exposure
Ratio of Apportioned Wealth Tax to Revenue



Notes: Wealth taxes are allocated to treated Norwegian firms in proportion to ownership-weighted revenues. The allocation weight is i's ownership share in f times f's revenues, divided by the corresponding sum across all firms owned by i, including f. Taxes are calculated from reported taxable wealth where available and imputed wealth otherwise, then scaled up by $1/(1 - \text{dividend tax rate})$. Appendix Figure OA.8 compares this measure with taxes calculated only from reported taxable wealth.

Wealth taxes are allocated to firms in proportion to ownership-weighted revenues across each owner's portfolio. For example, an owner with 50% stakes in two firms with revenues of 50 and 150 allocates 1/4 of their wealth tax to the first firm and 3/4 to the second. The procedure uses taxes calculated from reported taxable wealth where available and imputed taxable wealth otherwise (see Appendix B.2).

Figure 13 demonstrates the effects on wealth tax exposure. Wealth tax exposure is measured as allocated wealth tax payments (scaled up to account for implied dividend tax payments) divided by the firm's total revenues. By 2024, exposure has increased by about 1.6 percentage points relative to 2020. Appendix Figure OA.8 compares this measure with taxes calculated only from reported taxable wealth. The latter also restricts treated firms to those for which owners with reported taxable wealth hold more than 50% of the shares.

4.5 Effect on firm payouts

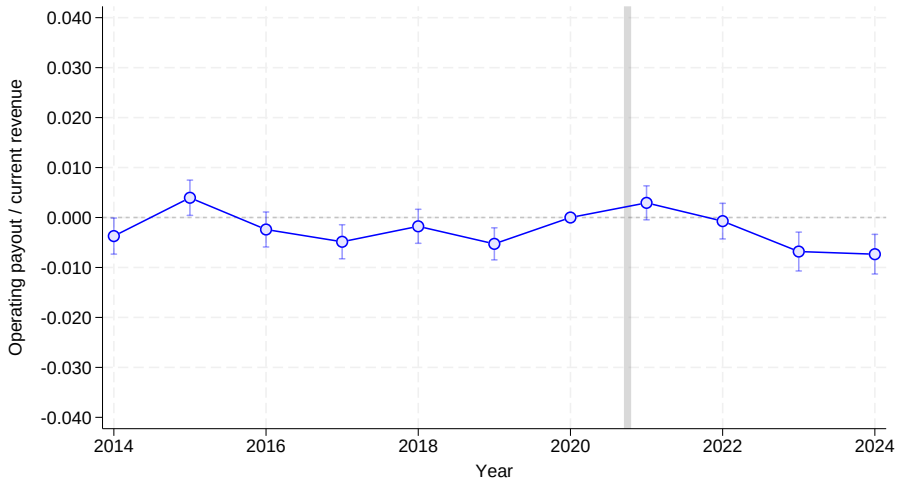
The key challenge in studying the effect on dividend payments is that wealthy individuals typically own corporations through several layers of ownership. If an owner owns firm A through B and C, a dividend that originates in firm A and passes through B and C would be observed three times in the data. To avoid this double (and triple) counting, I define "operating

dividends" as

$$\text{OperatingDividends}_{f,t} \equiv \text{NetIncome}_{f,t} - \text{FinancialIncome}_{f,t} - \Delta\text{RetainedEarnings}_{f,t}. \quad (7)$$

With this definition, which excludes *financial income* from net income, I do not count dividends that are received by f and immediately paid back out.

FIGURE 14: NO *INCREASE* IN PAYOUTS



Notes: Operating dividends equal net income minus financial income and the change in retained earnings. These dividends are typically paid out in $t+1$. Operating dividends are divided by concurrent revenues and then winsorized at the 5% and 95% levels. This measure avoids counting dividends received and passed on through several ownership layers (see text for additional discussion). Appendix Figure OA.16 considers all three settings (domestic, foreign firms, Nordic cross-country setting.)

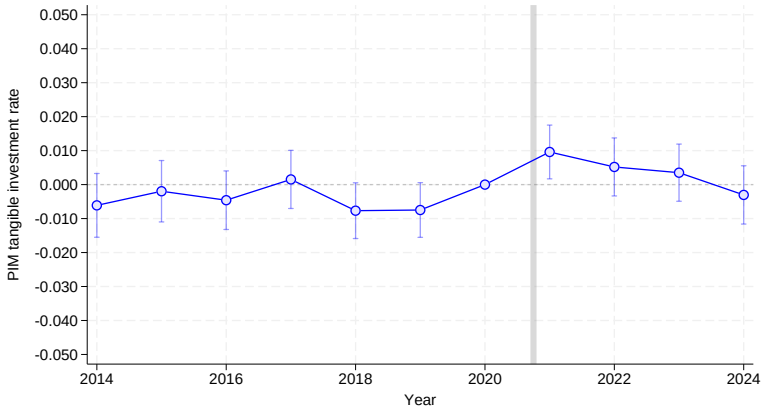
I report the results in Figure 14. We see a spike in dividend payments in 2015, which is undoubtedly attributable to anticipatory responses to dividend tax increases during 2016–2019. We further see a modest spike in 2021, which is likely driven by anticipatory responses to higher dividend taxes starting in 2022. However, we see no evidence of higher dividend payments during 2022–2023 during which wealth tax payments were substantially higher. During 2023–2024, payout rates are in fact lower for Norwegian-owned firms, which is consistent with low-EIS behavior in which wealth taxation may increase saving. Appendix Figure OA.16 shows results for the two additional samples. In the foreign-firm sample (contrasting Norwegian- and non-Norwegian-owned firms), I find no effect on payouts. In the cross-country sample, effects are similar in that they feature a reduction in payouts.

These findings do not imply that wealth taxes are not financed by dividends. Business owners may still extract dividends from their holding companies that do not have operating revenues and are thus dropped from my main regressions (as in, e.g., Figure 13). Consistent with this, Appendix Figure OA.10 shows that direct dividends to owners are considerably higher than the dividend allocated from the firms in my sample. This is especially true for 2021, which features anticipatory responses to higher dividend taxes, and for the 2022–2024 period. My

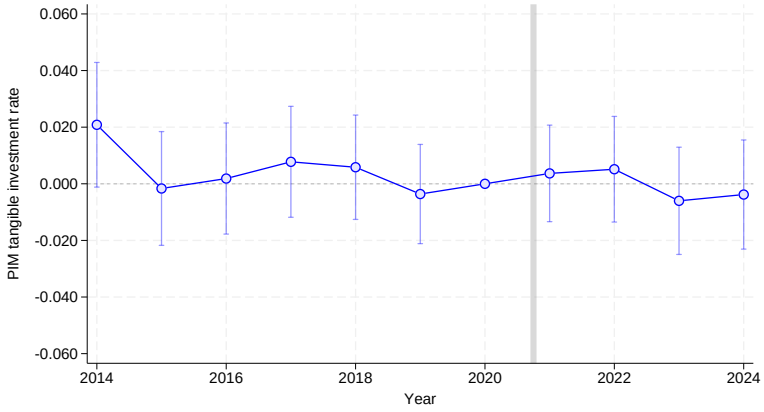
findings instead suggest that wealth taxes are not, on average, financed by extracting capital from operating companies.

FIGURE 15: NO REDUCTION IN INVESTMENT RATES

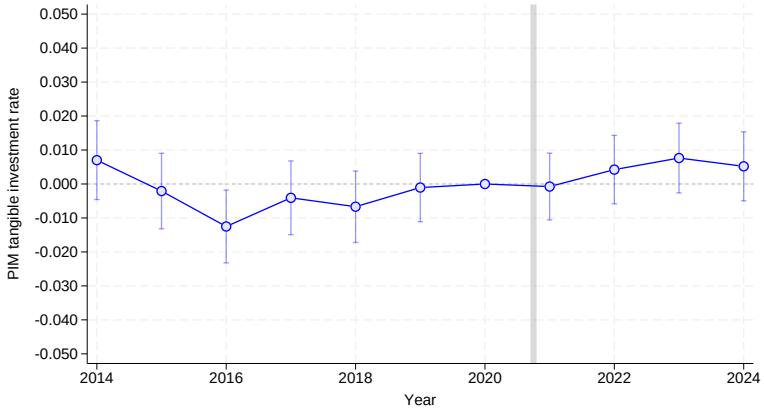
Panel A: Norwegian-domiciled firms



Panel B: Foreign-domiciled firms



Panel C: Nordic cross-country sample



Notes: Investment is defined as $I_{f,t} \equiv \Delta TFA_{f,t} + DEP_{f,t}$, which is then divided by the average of current and lagged capital stocks constructed using the perpetual inventory method (PIM). The resulting ratio is winsorized at the 5% and 95% levels.

4.6 Corporate investment

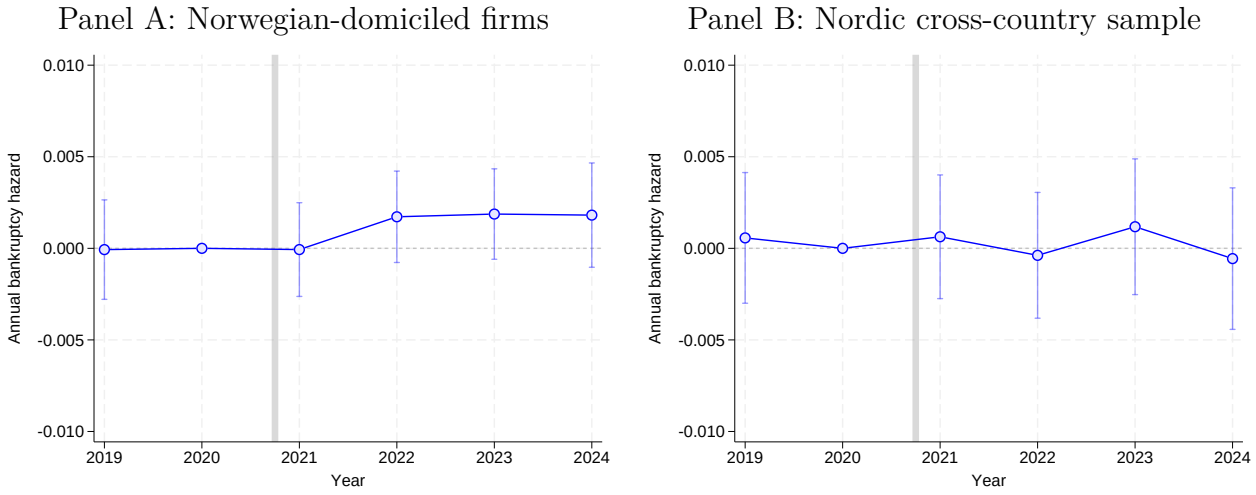
Figure 15 considers the effect on corporate investment. I measure investment as $I_{f,t} \equiv \Delta TFA_{f,t} + DEP_{f,t}$, where TFA is tangible fixed assets and DEP is depreciation. Investment is divided by the average of current and lagged capital stocks constructed using the perpetual inventory method (PIM).

There is no indication that higher wealth tax payments during 2022–2024 coincided with less investment, consistent with a low-to-moderate EIS among business owners (see Table 2). In the absence of financial frictions, where individuals can save and borrow at the risk-free rate, the model implies that the reforms increase the log user cost of capital by 0.0443.[17] Applying the mid-point of the estimates in Chodorow-Reich (2025) then implies a reduction in the investment rate of 2.2 percentage points. In contrast, the lower bounds of the 95% confidence intervals during 2022–2024 are all above −1.2 percentage points.

Panels B and C show similar results for foreign-domiciled firms and in the Nordic cross-country sample. Appendix Figure OA.11 considers the investment, $I_{f,t}$, scaled by revenue, in the domestic sample, and also finds no effect. Appendix Figure OA.12 shows that the results do not change if I drop firms whose largest owner had outmigrated by May 2024.

4.7 Firm bankruptcy

FIGURE 16: WHETHER THE REFORM INDUCED CORPORATE BANKRUPTCY



Notes: The outcome is entry into bankruptcy during year t, using Orbis status histories as of February 2026. Firms must have filed accounts in $t-2$. Observations after bankruptcy or other exits, including dissolution, are excluded. Panel A uses the domestic sample. Panel B compares Norwegian firms with Swedish and Finnish firms. Regressions include a treatment indicator and controls interacted with year indicators, but no firm fixed effects. The omitted year is 2020.

[17] When individuals can save and borrow at the same rate, the implied effect does not depend on the EIS and is given by the final row of Table 2, which assumes that the owner saves at every date.

Figure 16 examines whether Norwegian-owned firms were more likely to file for bankruptcy after the reforms. I consider both the domestic and Nordic cross-country samples. In neither case is there a significant effect on the propensity of Norwegian-owned firms to enter bankruptcy. In both cases, the 95% confidence intervals preclude increases in the annual hazard rate above 0.5 percentage points.

4.8 Robustness and heterogeneity

Are there larger effects in subsamples potentially more affected by the reforms? Appendix Figure OA.15 retains only treated firms with an above-median ratio of summed allocated wealth tax to summed revenues during 2016–2018, requiring observations in all three years. All control firms are retained. I show that this leads to about a 50% larger effect on wealth tax exposure (allocated wealth tax divided by revenue). However, I still find no effect on the investment rate, and the effect on payouts is still negative in 2023 and 2024.

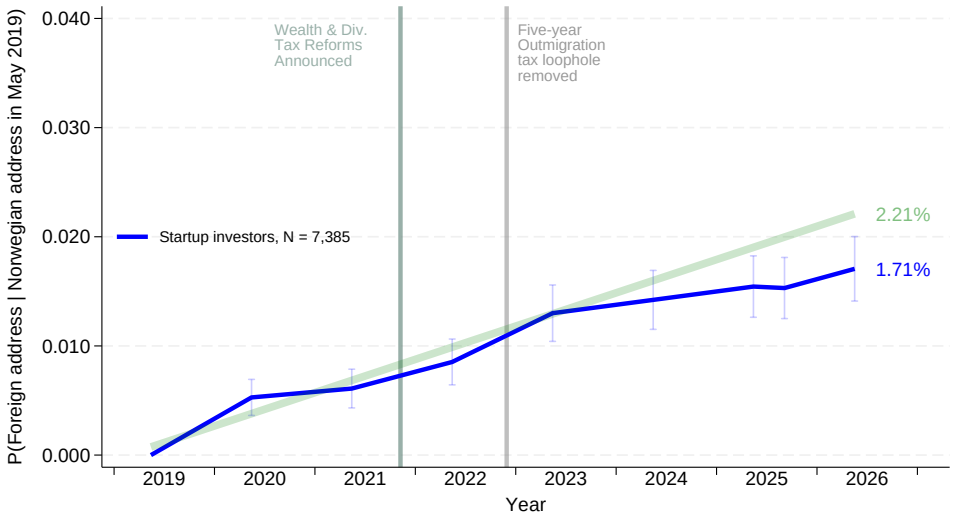
5 Disruptions to entrepreneurial finance

Whether startup investors outmigrated at higher rates after the reforms. A prominent literature in financial economics emphasizes the importance of well-functioning local capital markets in facilitating entrepreneurship (Guiso, Sapienza, and Zingales 2004; Kerr and Nanda 2009; Samila and Sorenson 2011). Several papers emphasize how physical distance between financiers and firms may exacerbate financial frictions (Lerner 1995; Petersen and Rajan 2002; Bernstein, Giroud, and Townsend 2016). Hence, I examine whether there were excess outmigration rates of *startup investors* after the 2021–2024 reforms.

Figure 17 plots cumulative outmigration rates of the subset of business owners who were investors in startups as of 2018. I define startups as firms awarded grants by Innovation Norway. Interestingly, there is no evidence of excess outmigration among these investors. If anything, these investors appear to outmigrate at lower than counterfactual rates following the closure of the five-year loophole.

These results do not imply that no startup financiers left the country due to the reform. Appendix Figure OA.13 focuses on individuals in the top 1% of business owners. The findings show (imprecisely) that those who outmigrated between the spring of 2022 and the spring of 2024 were equally likely to be startup investors (at rates of about 15–20%) as nonmigrants. Nevertheless, on average among startup investors, there was no tendency of increased outmigration rates.

Figure 17: Outmigration Rates of Startup Investors



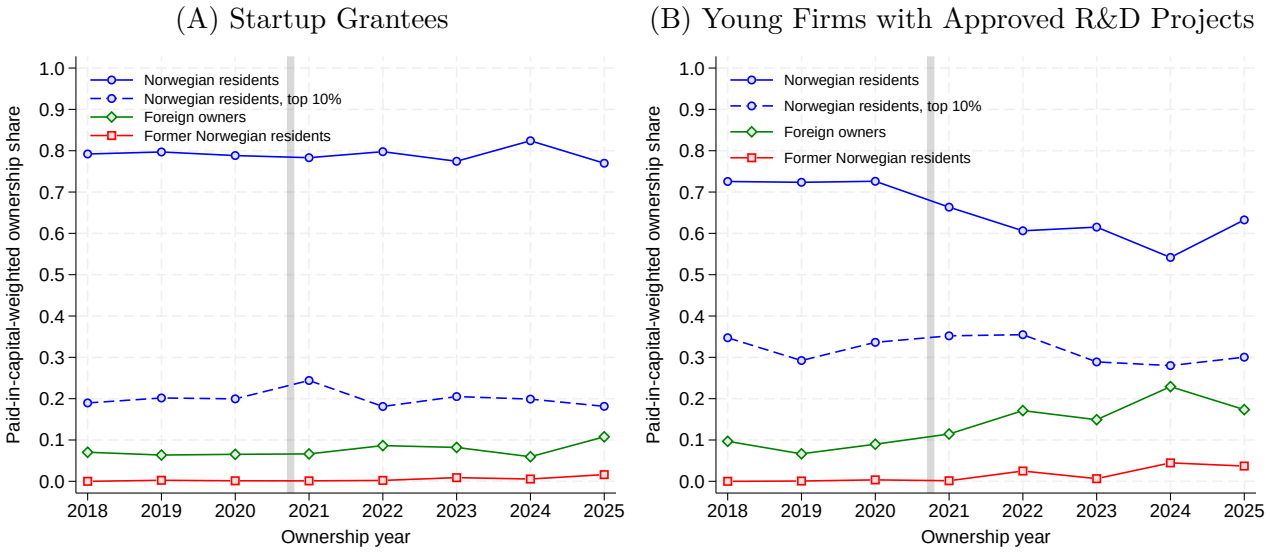
Notes: This figure considers the share residing abroad among startup investors residing in Norway in the 2018 shareholder register (measured in May 2019). I classify an individual as a startup investor if they directly or indirectly own at least 1% of a firm in 2018 that received a startup grant from Innovation Norway during 2015, 2016, 2017 or 2018. I drop individuals whose smallest stake in a startup is $\geq 50\%$ to limit the inclusion of owner-manager entrepreneurs. The shaded solid line is the extrapolated pre-trend. See Figure 10 for details regarding the methodology.

Whether Norwegian investors invested less in startups. If the 2021–2024 reforms reduced Norwegian investors' incentives or ability to invest in high-potential new firms, the ownership composition of startups should change. Norwegians' ownership share should then shrink after the reforms, in particular if foreign investors continue to finance these firms or step in to replace Norwegian financing. I define innovative startups in two ways (Panels A and B in Figure 18). For both definitions, I consider only firms with some ownership ($\geq 10\%$) attributable to Norwegian residents to avoid counting fully-owned subsidiaries of foreign firms.

First, I consider first-time recipients of startup grants from Innovation Norway. I find no evidence that the capital-weighted share of Norwegian ownership decreases over time. Appendix Figure OA.17 demonstrates that there is also no decrease when considering unweighted ownership shares.

Second, Panel B considers young firms who are eligible to receive R&D tax credits via the *SkatteFUNN* program. Here, I find a decline in the ownership share of Norwegian residents. The share attributable to top-10% business owners also decreases from 2023, while the share of foreign ownership increases. This is consistent with reduced investment from Norwegian investors, and the effect appears to be driven by high-capital investments. When considering unweighted ownership shares, the share attributable to top-10% business owners does not decline.

FIGURE 18: OWNERSHIP COMPOSITION OF INNOVATIVE STARTUPS



Notes: This figure considers the ownership composition of two sets of startups over time. Panel A considers firms awarded startup grants from Innovation Norway. I count ownership at year t when the startup was granted an award in year $t-2$. I only count a startup's first observed award. Innovation Norway is a government-owned agency, and the startup grants are intended to finance the growth of high potential, innovative new firms. Panel B considers firms that are three years old and had an approved R&D project with *SkatteFUNN* covering year $t-1$ or $t-2$. For Panels A and B: Ownership is weighted by ownership shares times the startup's paid-in capital, where paid-in-capital is winsorized at the 95th percentile each year. The sample of startups is limited to firm-year observations in which at least 10% is ultimately owned by Norwegian residents (e.g., no fully-owned subsidiaries of foreign corporations). Average sample sizes (number of startups) are 458 in Panel A and 291 in Panel B. See Appendix Figure OA.17 for the unweighted (ownership not weighted by paid-in-capital) versions of these plots

6 Discussion

This paper studies the controversial 2021–2024 Norwegian tax reforms that increased the tax burden of business owners. While I document clear effects on outmigration, these effects are confined to about 100 wealthy individuals who outmigrated during 2022–2023. There is little evidence of outmigration or reduced inmigration of other entrepreneurs or inventors, and there is little evidence to suggest that the wealth tax reduced corporate investment in the short run.

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Online Appendix

A User cost of capital, model and calibration details

A.1 Environment and household problem

The owner begins at date 0 with \$1 of retained earnings and chooses $k \in (0,1)$ to retain in the firm. The remaining $1-k$ is distributed immediately. Because the initial dollar consists of retained earnings, both this date-0 distribution and the capital distributed at liquidation are subject to dividend tax. The firm operates for $T \geq 2$ years and is liquidated at the end of year T. Retirement lasts $J \geq 2$ years, beginning at date T.

The firm earns annual operating income $F(k)$ before depreciation and corporate income tax, where $F(0) = 0$ and F is increasing and strictly concave. Capital depreciates at rate δ. The firm replaces depreciation each year but cannot otherwise adjust its capital stock or hold financial assets before liquidation. Define $q(k)$ so that $(1-\tau^{Div})q(k)$ is the owner's annual cash flow from the firm, net of corporate, dividend, and personal wealth taxes:

$$q(k) \equiv (1-\tau^{CIT})[F(k)-\delta k] - \frac{\theta^S \tau^w}{1-\tau^{Div}} k. \tag{OA.1}$$

Let $s_{t+1} \geq 0$ denote the amount saved personally at date t. Personal saving earns r^{save} after personal income tax and has wealth-tax assessment fraction θ^P, so one dollar saved at date t yields $1 + r^{save} - \theta^P \tau^w$ at date $t+1$. The owner cannot borrow. Consumption satisfies

$$\begin{aligned}
c_0 &= (1-\tau^{Div})(1-k) - s_1, \\
c_t &= (1-\tau^{Div})q(k) + s_t(1 + r^{save} - \theta^P\tau^w) - s_{t+1}, && t = 1, \ldots, T-1, \\
c_T &= (1-\tau^{Div})[k + q(k)] + s_T(1 + r^{save} - \theta^P\tau^w) - s_{T+1}, \\
c_t &= s_t(1 + r^{save} - \theta^P\tau^w) - s_{t+1}, && t = T+1, \ldots, T+J-2, \\
c_{T+J-1} &= s_{T+J-1}(1 + r^{save} - \theta^P\tau^w).
\end{aligned} \tag{OA.2}$$

Let $\beta \in (0,1)$ denote the discount factor, and let u be increasing, twice differentiable, and strictly concave. Choosing $k \in (0,1)$ and $s_{t+1} \geq 0$ for $t = 0, \ldots, T+J-2$, the owner solves

$$\max_{k, \{s_{t+1}\}} \sum_{t=0}^{T+J-1} \beta^t u(c_t) \quad \text{subject to equation (OA.2).} \tag{OA.3}$$

A.2 Required return

Let R denote the annual marginal revenue product that makes the owner willing to retain k in the firm. A marginal increase in k reduces the date-0 distribution by one dollar. In each operating year, it raises $q(k)$ by

$$(1-\tau^{CIT})(R-\delta)-\frac{\theta^S\tau^w}{1-\tau^{Div}},$$

and it returns one dollar of capital at liquidation. The interior first-order condition for k is

$$\begin{aligned} u'(c_0) = &\sum_{t=1}^{T-1}\beta^t u'(c_t)\left[(1-\tau^{CIT})(R-\delta)-\frac{\theta^S\tau^w}{1-\tau^{Div}}\right] \\ &+\beta^T u'(c_T)\left[1+(1-\tau^{CIT})(R-\delta)-\frac{\theta^S\tau^w}{1-\tau^{Div}}\right]. \end{aligned} \tag{OA.4}$$

The left-hand side is the marginal-utility cost of retaining one additional dollar at date 0. The right-hand side is the discounted marginal-utility value of the resulting operating cash flows and liquidation proceeds. Rearranging gives equation (3).

At the pre-reform optimum, $F'(k)=R$. The fixed-k comparison evaluates equation (3) under the post-reform tax system at the pre-reform capital stock. It does not impose $R=F'(k)$ after the tax change.

When $s_t>0$ for every $t=1,\dots,T$, the Euler equations

$$u'(c_{t-1})=\beta(1+r^{save}-\theta^P\tau^w)u'(c_t), \qquad t=1,\dots,T, \tag{OA.5}$$

reduce the required return to equation (4). If $s_t=0$ at any date, the Euler condition is an inequality, and the required return also depends on the consumption profile in equation (3).

A.3 EIS comparative static

For the EIS comparative static, suppose that $s_t=0$ for every $t=1,\dots,T$. The owner continues to choose nonnegative personal saving during retirement. Suppose utility is isoelastic, with $u'(c)=c^{-1/\sigma}$, where σ is the EIS. During the business phase,

$$c_0=(1-\tau^{Div})(1-k), \qquad c_t=(1-\tau^{Div})q(k), \quad t=1,\dots,T-1. \tag{OA.6}$$

At date T, liquidation pays $(1-\tau^{Div})[k+q(k)]$ into the owner's retirement budget. The marginal value of an additional dollar at date T is $u'(c_T)$.

At the pre-reform optimum, define

$$\sigma^*_{T,J} \equiv \frac{\sum_{t=1}^{T-1} \beta^t u'(c_t)\frac{kq_k}{q} + \beta^T u'(c_T)\frac{k(1+q_k)}{k+q}}{\sum_{t=1}^{T-1} \beta^t u'(c_t) + \beta^T u'(c_T)}, \qquad q_k = (1-\tau^{CIT})[F'(k)-\delta] - \frac{\theta^S \tau^w}{1-\tau^{Div}}. \quad \text{(OA.7)}$$

The ratios kq_k/q and $k(1+q_k)/(k+q)$ are the elasticities of annual business cash flow and liquidation proceeds with respect to k. The cutoff $\sigma^*_{T,J}$ is their marginal-utility-weighted average.

For an increase in θ^S, holding k, τ^w, τ^{Div}, and θ^P fixed, differentiating equation (OA.4) gives equation (5). The required marginal product falls when $\sigma < \sigma^*_{T,J}$ and rises when $\sigma > \sigma^*_{T,J}$. If $q > 0$ and $q_k > 0$, $F(0) = 0$ and strict concavity of F imply that both elasticities in equation (OA.7) lie between zero and one. It follows that $0 < \sigma^*_{T,J} < 1$.

This derivative changes only θ^S. The calibration instead changes θ^S, τ^w, and τ^{Div} jointly and reoptimizes the consumption and saving path.

A.4 Calibration

Panel A of Table 2 reports the parameters used to calibrate the model.

The baseline required return satisfies

$$(1-\tau^{CIT})(R_0-\delta) - \frac{\theta^{S,pre}\tau^{w,pre}}{1-\tau^{Div,pre}} = r^{save} - \theta^P \tau^{w,pre} + \pi. \quad \text{(OA.8)}$$

With the parameter values in panel A of Table 2,

$$\begin{aligned} R_0 &= 0.10 + \frac{0.03 - 0.0085 + 0.02 + \frac{0.55(0.0085)}{1-0.3168}}{1-0.22} \\ &= 0.16198. \end{aligned} \quad \text{(OA.9)}$$

Dividend tax applies to both the foregone date-0 distribution and subsequent business payouts. It enters this comparison only through the wealth-tax gross-up in equation (OA.8).

For each EIS, I jointly choose A such that equation (3), evaluated under the pre-reform tax system, satisfies

$$R^{pre} = R_0, \qquad F'(k) = R^{pre}. \quad \text{(OA.10)}$$

I then hold A and k fixed, impose the changes in θ^S, τ^w, and τ^{Div}, reoptimize the consumption and saving path, and reevaluate equation (3). Table OA.1 reports the optimal business-phase

personal-saving paths.

TABLE OA.1: OPTIMAL PERSONAL-SAVING PATHS

EIS	Pre-reform path	Post-reform path
0.20	Zero at 1, positive at 2–20	Positive at 1–7, zero at 8–20
0.25	Zero at 1, positive at 2–20	Positive at 1–5, zero at 6–20
0.50	Zero at 1–2, positive at 3–20	Zero at 1–20
1.00	Zero at 1–12, positive at 13–20	Zero at 1–20
2.00	Zero at 1–19, positive at 20	Zero at 1–20
6.00	Zero at 1–20	Zero at 1–20

Notes: This table summarizes the personal-saving paths for different values of the EIS. Each entry reports whether s_t is zero or positive. For example, "zero at 1" means $s_1 = 0$, so the no-borrowing constraint binds for the first saving choice.

Partial assessment of personal saving. As a robustness exercise, set $\theta^P = 0.5$. Equation (OA.8) then gives

$$R_0^{half} = 0.10 + \frac{0.03 - 0.5(0.0085) + 0.02 + \dfrac{0.55(0.0085)}{1 - 0.3168}}{0.78} = 0.16743. \tag{OA.11}$$

I recalibrate A and k to this baseline for each EIS and repeat the fixed-k comparison (Table OA.2).

TABLE OA.2: PARTIAL ASSESSMENT OF PERSONAL SAVING

EIS	Baseline k	R^{post}	$100 \times \Delta \log R$	$\Delta(I/K)$ (pp.)
0.20	0.9358	0.1496	−11.285	5.642
0.25	0.9366	0.1503	−10.782	5.391
0.50	0.9405	0.1628	−2.813	1.406
1.00	0.9476	0.1697	1.373	−0.687
2.00	0.9596	0.1735	3.537	−1.769
6.00	0.9860	0.1760	5.008	−2.504

The required-return change equals zero at an EIS of approximately 0.756. Personal saving is positive from date 2 onward in every pre-reform solution. After the reform, it is positive at dates 1–20 for an EIS of 0.2, at dates 2–20 for an EIS of 0.25 and 0.5, at dates 11–20 for an EIS of 1, at dates 19–20 for an EIS of 2, and at no date for an EIS of 6. Relative to the benchmark, partial assessment makes the negative required-return changes larger in absolute value at low EIS and the positive required-return changes smaller at high EIS.

A.5 Dividend taxes and rate-of-return deduction

In the current tax system ("aksjonærmodellen"), dividends paid from a given firm to a given individual are tax-exempt up to a threshold approximately equal to the tax-authority-determined risk-free rate times the shares' adjusted cost basis ("skjermingsfradrag"). The cost basis is adjusted in that it increases with previous unused exemptions. This implies that even though some portion of dividends meant to finance wealth tax payments may not trigger an immediate dividend tax liability, it will nevertheless have an equivalent tax cost by reducing the adjusted cost basis.

Consider 2024. The dividend tax rate is 37.84%, the top marginal tax rate on book value business wealth is 0.88%, and the relevant risk-free rate is 3.9%. Further consider a simple case in which the tax value of a firm equals the book value of equity. If the (adjusted) cost basis equals the value of equity which equals \$100. In that case, the wealth tax payment is \$0.88 but \$3.9 of dividends may be paid out without incurring additional *immediate* dividend tax. However, in 2022, the adjusted cost basis equaled only 18% of the pre-stock-discount assessed value of Norwegians' personally-held unlisted stocks (Grindaker and Vestad, 2025, Figure 4.3). Applying this 18% to 2024, suggests an effective exemption rate on tax-value equity of 0.7% ($18\% \times 3.9\%$). Thus dividend payments to finance the wealth tax (0.88%) can only be partially exempt from dividend taxation, and only to the extent that the owner would not already make use of the exemption for dividends for other purposes, e.g., financing consumption.

B Data appendix

B.1 AI

AI tools were used to draft, fix, and review various aspects of the code used to conduct this research.

B.2 Imputed tax payments for business owners

I first estimate

$$\log(TNW_{i,t}) = \beta_t \log(DirectBookEquity_{i,t}) + \alpha_t + \epsilon, \tag{OA.12}$$

on tax-list observations during 2012–2024 with positive taxable wealth and $DirectBookEquity_{i,t} > 1,000,000$, in municipality-years with at least 100 reported wealth observations. $DirectBookEquity_{i,t}$ is the sum of directly-held book equity allocated (via ownership share) to a given individual, with negative firm equity set to zero. The regression is run at the individual-year level, with

year-specific intercepts and slopes.

I then predict $\log(TNW_{i,t})$ for all i such that $DirectBookEquity_{i,t} > 100,000$. Predicted TNW is then set to $\exp(\widehat{\log(TNW_{i,t})} + 0.5\text{RMSE}^2)$, where $RMSE$ is the pooled root mean squared error of the regression above. I then use predicted TNW to calculate tax payments using the prevailing tax scheme for year t. I only impute taxes for Norwegian residents, measured during 2014–2024 as having a non-missing postcode in the appropriate shareholder register vintage. Taxes calculated from reported taxable wealth are used where available.

B.3 Administrative microdata.no data

Vintage. All results use `no.ssb.fdb:55`, which covers tax years through 2024 and residency status through 1 January 2026.

Sample and wealth groups. The sample is restricted to adults resident in Norway, using residency status on 1 January (`BEFOLKNING_STATUSKODE`) and month of birth (`BEFOLKNING_FOEDSELS_AAR_MND`). Annual migration rates cover 1994–2025, the cumulative figures follow those resident on 1 January 2018, and the outmigrant characteristics cover 2016–2025. Wealth groups are assigned using taxable net wealth (`SKATT_NETTOFORMUE`) assessed on 31 December of the preceding year, hence wealth is measured prior to the migration decision. Groups are constructed by taking the top decile, then the top decile of that group, and so on down to the top 0.01%. Figures report the non-overlapping bands these cuts imply (90–99%, 99–99.9%, and so on). Individuals without a wealth record are assigned zero taxable wealth, hence the groups refer to shares of the adult population.

Outmigration. An individual is counted as having emigrated during year t if resident at the start of year t and recorded as emigrated at the start of year $t + 1$. Deaths are kept as a separate category. For the cumulative figures, moves are dated within the year using address spells (`BOSATTEFDT_BOSTED`), taking the end of the last domestic spell as the date of the move.

Untaxed wealth. One set of cumulative figures sorts the 2018 cohort on untaxed wealth in place of taxable net wealth. Untaxed wealth is the end-2017 stock of shares and other securities (`INNTEKT_VERDIPAPIR`) and share savings accounts (`SKATT_ASK`), less realized capital gains (`INNTEKT_REALISASJONSGEVINST`) and dividends (`INNTEKT_AKSJEUTBYTTE`) reported over 2006–2017. It approximates the part of an owner's equity that has yet to be taxed at realization.

Revenue. Wealth tax revenue is measured as assessed wealth tax (SKATT_FORMUESKATT), and income tax revenue as assessed income tax.

Outmigrant characteristics. Those who emigrate during year t are compared with those who remain resident, both measured at 31 December of year $t-1$. Financial assets are shares and other securities at estimated market value (INNTEKT_VERDIPAPIR), and logarithms are taken among those with a positive holding. Statistics Norway restates this variable to market value from income year 2017 onward, hence the two earliest years rest on assessed values. Whether an individual holds foreign financial assets is determined by INNTEKT_FORMUE_UTLAND > 0, which covers bank deposits and securities held outside Norway. Country of birth is taken from BEFOLKNING_FODELAND and age from BEFOLKNING_FOEDSELS_AAR_MND.

Turnout. Turnout is taken from the electoral register (VALG_STEMT_SSB) for the 2017, 2021 and 2025 parliamentary elections, and reported across the wealth distribution.

Misc. The platform does not return statistics for very small groups, and it perturbs tabulated counts by ± 10. Group means and their standard errors are obtained from regressions on group indicators. The resulting statistics, such as the average migration rate within a group, are not perturbed. Standard errors are heteroskedasticity-robust, and confidence intervals on migration rates are Wilson intervals.

B.4 Firm samples and empirical specifications

Ownership links. Norwegian firms are linked between Orbis and the shareholder register by organization number. Direct ownership is measured as a fraction of outstanding shares. Indirect shares are the product of shares along each ownership chain, summed across chains reaching the same owner. Iteration follows all direct links and cumulative shares of at least 1%, stopping when a firm reappears in a chain or after 100 layers. Firm treatment status is fixed using 2018 ownership.

Norwegian ownership. In Settings 1 and 3, Norwegian firms are treated if individuals resident in Norway collectively own at least 67% of the firm, directly or indirectly. Residence is measured using the shareholder-register address. Domestic controls must appear in the 2018 shareholder register and satisfy the ownership restrictions in the main text. Ownership held through domestic partnerships counts toward the 33% limit for controls. The treatment threshold uses ownership attributed to identified individuals. The control group includes firms with partially unidentified owners.

Ownership of foreign firms. Setting 2 combines three sources of ownership information from Orbis in 2018. A firm is Norwegian-owned if Orbis identifies a Norwegian individual as its majority global ultimate owner, if its Norwegian corporate majority ultimate owner is more than 50% owned by Norwegian individuals, or if reported shareholder percentages attribute more than 50% of the firm to Norwegian individuals. For the latter calculation, shares held by Norwegian corporations are multiplied by the fraction of those corporations owned by Norwegian individuals, using the shareholder register. The calculations use Orbis' global ultimate owner (GUO50, GUO50C) and first-level shareholder (SHH, ISH) records. Each shareholder enters once, and the direct-share calculation requires consistent reported percentages. Controls must have an owner country in the selected group, with agreement between the GUO50 and GUO50C records. Firms with a Norwegian ultimate owner or any Norwegian direct shareholder recorded during 2016–2019 are excluded from controls.

Domestic sample. Setting 1 excludes finance, insurance, and real estate (NACE 64–66 and 68). Firms must report revenue in every year 2016–2019. The early size groups require positive real revenue in both 2013 and 2014, observed tangible assets in 2014, and a known incorporation year no later than 2014. Firms must belong to either a four-digit industry containing treated and control firms or an industry-and-size group containing both.[18] Revenue deciles are calculated on this sample using mean log real revenue in 2013–2014. The final sample retains three-digit industries containing both treated and control firms.

Foreign sample. Setting 2 includes firms in 27 European countries and controls owned from 14 Western European countries.[19] Finance, insurance, and real estate are excluded. Firms must report total assets in every year 2016–2019. Each operating-country-by-two-digit-industry group must contain treated and control firms.

Nordic sample. Setting 3 compares Norwegian firms satisfying the 67% ownership threshold with Swedish and Finnish firms. Denmark is excluded because of limited revenue and depreciation coverage in Orbis and the absence of a comparable Danish national-accounts depreciation series in the PIM inputs. The sample includes manufacturing industries NACE 10–17 and 20–32, excluding fish processing (10.2). Swedish and Finnish firms are excluded if Orbis records

[18] The latter groups combine two-digit industry, predicted 2015 log real revenue in half-log-point intervals, and the 2014 tangible-asset category. Predictions use a firm-specific linear trend over available positive revenues during 2010–2014. Firms qualifying through this prediction-based grouping require at least three observations. Trend slopes are winsorized at the pooled 5th and 95th percentiles before prediction.

[19] Operating countries are Austria, Belgium, Bulgaria, Croatia, the Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, the Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, and the United Kingdom. Control-owner countries are Austria, Belgium, Denmark, Finland, France, Germany, Greece, Iceland, Italy, the Netherlands, Portugal, Spain, Sweden, and the United Kingdom.

a Norwegian majority ultimate owner or more than 50% Norwegian direct ownership in 2018. Controls include firms with missing ownership information. Controls with an observed incorporation year after 2018 are excluded. Three-digit industries must contain Norwegian and control firms.

Fixed effects. Regressions for continuous outcomes include firm fixed effects and industry-by-year fixed effects, using three-digit NACE in Settings 1 and 2 and four-digit NACE in Setting 3. Setting 2 also includes operating-country-by-year fixed effects. All settings include year interactions with firm age in 2014, grouped into 0–4, 5–9, 10–18, and at least 19 years. Settings 2 and 3 retain a separate category for firms with unknown age or incorporated after 2014. Industry, age, and size effects enter additively. The estimation period is 2014–2024 and the omitted year is 2020. Firm-years have equal weight. Standard errors are clustered by firm.

Size controls. Setting 1 includes year interactions with the revenue deciles defined above and four 2014 tangible-asset categories: nonpositive assets, up to NOK 0.5 million, NOK 0.5–5 million, and above NOK 5 million. Assets are converted at fixed 2020 exchange rates. Setting 2 uses the same tangible-asset categories, with a separate category for missing assets, and groups based on total assets in 2013–2014.[20] Setting 3 includes year interactions with quartiles of mean log positive real revenue and total assets during 2012–2014. Available years are averaged, and missing values form separate categories.[21]

Bankruptcy sample and regression. Annual bankruptcy regressions cover 2019–2024. Firms must have filed unconsolidated accounts in $t-2$ and are followed through the year of bankruptcy or other exit. The domestic sample retains the early-data requirements described above, replacing the 2016–2019 revenue requirement with the filing requirement at $t-2$. The treatment indicator replaces firm fixed effects. Regressions also include treatment-by-year interactions and the setting's industry, size, and age controls interacted with year. Setting 3 additionally includes country fixed effects.[22] The omitted year is 2020 and standard errors are clustered by firm.

[20] The total-asset groups distinguish whether assets are observed in both years, only in 2013, only in 2014, or neither. When both are observed, groups are the integer part of $\log(1{,}000 + \max\{A_{2013}, 0\}) + \log(1{,}000 + \max\{A_{2014}, 0\})$, with assets converted to NOK at fixed 2020 exchange rates.

[21] Quartile cutoffs are calculated on Norwegian, Swedish, Danish, and Finnish manufacturing firms with at least two positive-asset years in 2012–2014 and three-digit industries containing both groups. These cutoffs are applied to firms with any observed early mean in the Norwegian, Swedish, and Finnish regression sample.

[22] I include an indicator for Finland, which, since only Norway, Sweden, and Finland are in the sample, amounts to country fixed effects.

B.5 Firm outcomes

Financial statements. Orbis Historical unconsolidated accounts (U1 and U2) are taken from the April 2026 vintage. The accounting year is the fiscal closing year. One statement is retained per firm-year, prioritizing the latest closing date and then the longest reporting period. Flows retain their reported amounts and reporting periods. Estimated revenue and employment are set to missing. Stock changes require consecutive years and the same reporting currency. Nordic accounting outcomes require local-currency reporting. Industries use NACE Rev. 2 from the December 2025 vintage. Each regression uses firm-years with the required outcome and control variables.

Operating dividends. Inferred total dividends equal net income minus the change in retained earnings. Operating dividends further subtract financial income, as defined in the main text.[23] In Norwegian accounts, dividends proposed in year t will enter operating dividends, by this definition, in year t. The domestic payout ratio requires positive current revenue. Its denominator is current revenue with a floor of NOK 100,000 in 2020 prices to avoid assigning large payout ratios to small absolute amounts.

Investment and capital. Book investment is the change in tangible fixed assets plus reported depreciation and amortization. Negative tangible assets are set to zero. Depreciation enters as reported, with missing values set to zero when current and lagged tangible assets and current revenue are observed. These rules are common to all three settings. Let $I^r_{f,t}$ denote investment adjusted for investment prices. PIM capital and the investment rate are

$$K_{f,t} = \max\{(1-\delta_{g,t})K_{f,t-1} + I^r_{f,t}, 0\}, \qquad \text{InvestmentRate}_{f,t} = \frac{I^r_{f,t}}{(K_{f,t}+K_{f,t-1})/2}.$$

The calculation uses available accounts from 2004 onward. Capital is initialized from observed tangible assets adjusted for investment prices, and reinitialized after gaps, missing lagged capital, or a currency change. The denominator must be positive. The alternative domestic book-investment-to-revenue ratio divides book investment by positive current revenue.

Prices, currencies, and depreciation rates. Settings 1 and 2 use Norges Bank's December 2020 exchange rates and Statistics Norway's industry-level investment-price indices. Depreciation rates are annual consumption of fixed capital divided by lagged net capital stock, using Norwegian industry-section series from the national accounts. In Setting 3, which features firms domiciled across Nordic countries, I use annual 2020 exchange rates from Eurostat and a

[23] Reported net income is replaced by profit after tax when profit after tax and extraordinary-and-other expenses are both reported and equal.

geometric-mean investment-price index (based on indices for Norway, Sweden, Denmark, and Finland). For depreciation rates, I use country-specific manufacturing consumption of fixed capital divided by lagged net capital stock. The Norwegian series come from Statistics Norway, and the Swedish and Finnish series from Eurostat.

Outliers. Payout-to-revenue ratios are bounded to $[-1, 1]$. Payout ratios, PIM investment rates, and book-investment-to-revenue ratios are then winsorized at the 5th and 95th percentiles. Cutoffs are calculated separately by outcome and setting, pooling treated and control observations during 2014–2020, and applied unchanged to subsequent years. Allocated wealth-tax-to-revenue ratios use a NOK 100,000 revenue floor in 2020 prices and are bounded to $[0, 1]$.

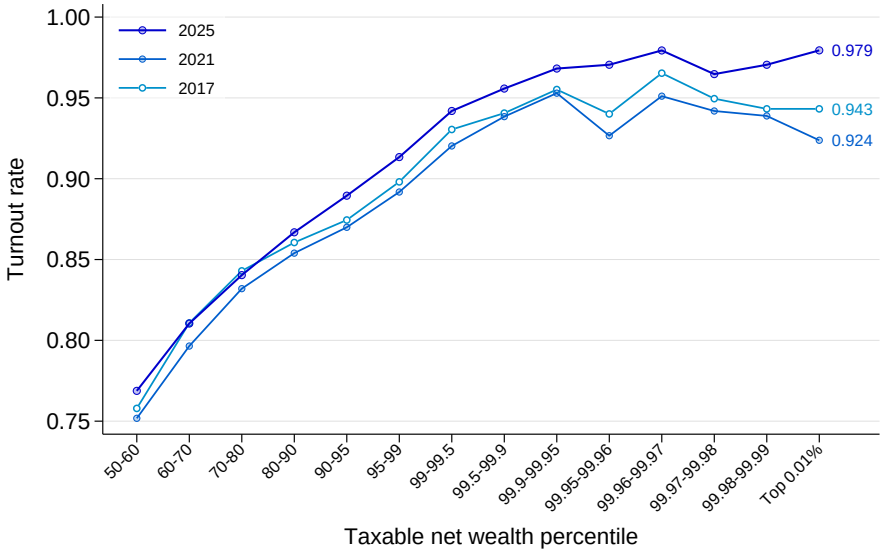
Bankruptcy status. I use the first dated Bankruptcy or Dissolved (bankruptcy) status observable in the status history of the firm in Orbis. Other exits include dissolution, inactivity, unknown status, and administrative suspension.[24] Firms with an undated bankruptcy record are excluded. Firms without a status record are retained as having no observed bankruptcy.

Supplementary financial statements. For the event-study analyses of outmigrants' firms, I use financial statement data from the business registries as of September 2026 rather than Orbis. As of this date, the business register data has nearly full coverage of 2025 financial statements whereas Orbis has about half coverage. This allows me to extend the analyses into 2025. When considering firm outcomes (regardless of outmigration), I use only the Orbis data to stay consistent across samples.

[24] The exact Orbis categories are Dissolved, Dissolved (demerger), Dissolved (liquidation), Dissolved (merger or take-over), Inactive (no precision), Status unknown, and Administratively suspended. Firms classified as In liquidation remain in the sample until a qualifying bankruptcy or exit.

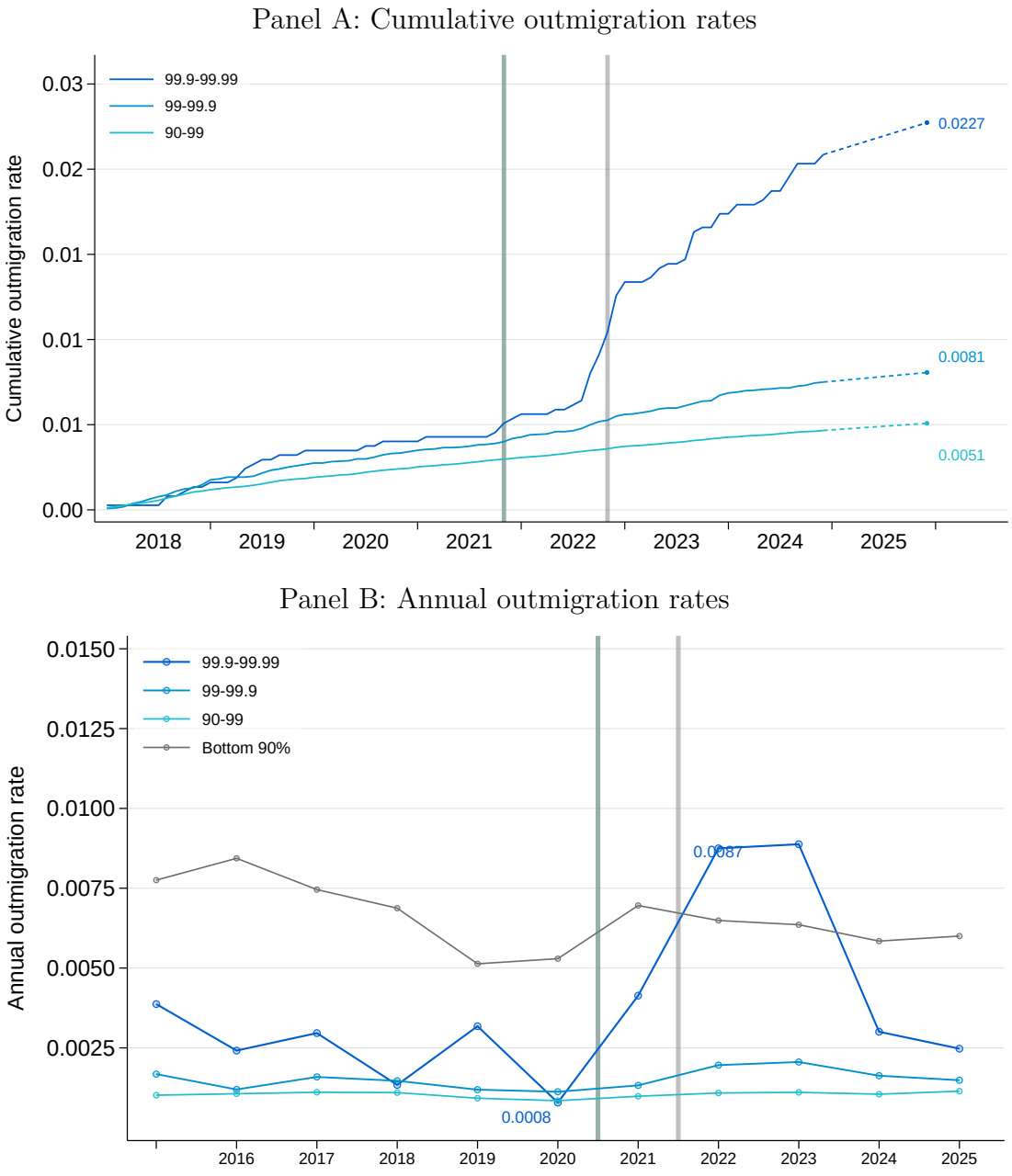
C Additional figures

Figure OA.1: Election Turn-out Increase Among the Ultra Wealthy



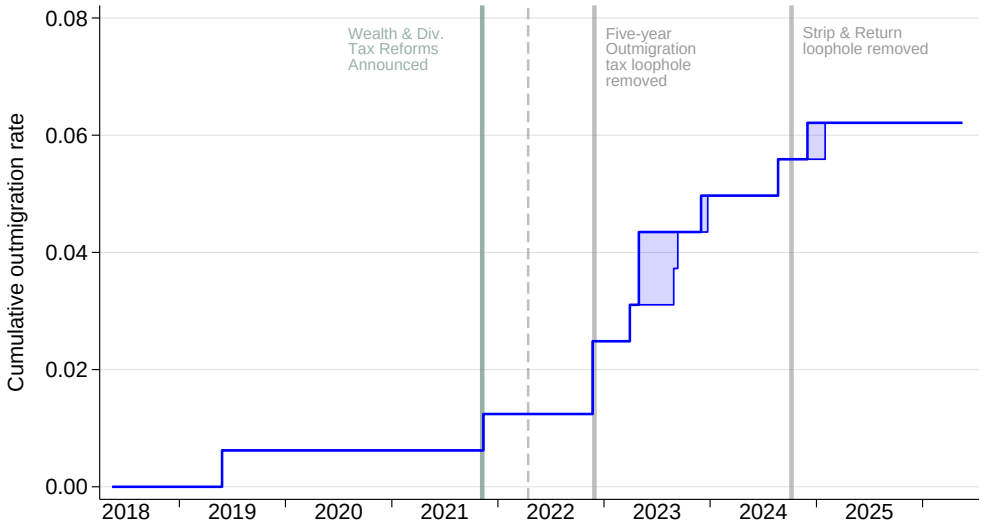
Notes: This figure shows voter participation (whether voted) across the taxable net wealth distribution for the 2017, 2021, and 2025 parliamentary elections. Data obtained from microdata.no.

FIGURE OA.2: OUTMIGRATION OF WEALTHY NORWEGIANS, ADMINISTRATIVE DATA. OMITTING THE TOP 0.01% FOR READABILITY



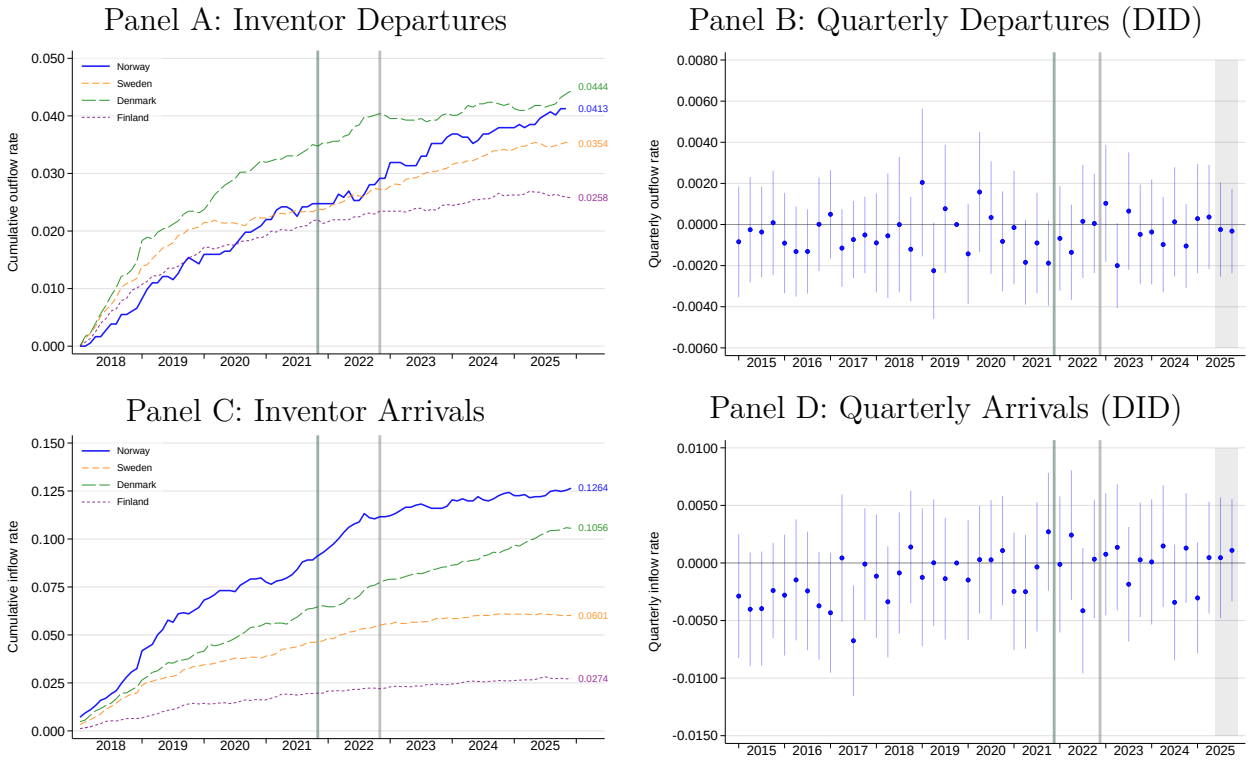
Notes: Figure 2 shows outmigration rates for different wealth groups, regardless of business ownership, around the 2021–2024 reforms. *This figure* omits the top 0.01% such that the migration rates for the other wealth bins become more distinguishable. Panel (A) shows cumulative outmigration rates since 2017. Sample consists of those resident 01/01/2018, and wealth groups are assigned using 2017 tax return data. I impute exact migration dates from address registers. If someone outmigrates during 2020, and their address changes in May 2020, I assign May 2020 as the date of their move. Panel (B) shows annual outmigration rates for a given year among those resident at the beginning of the year, and assigns wealth based on the prior year's tax data. The underlying data comes from microdata.no.

Figure OA.3: Outmigration of High-Profile Investors



Notes: This figure shows cumulative outmigration rates for high-profile investors: any named investor on the Finansavisen Business Journal's list of high-profile investors ("bellwethers") as of August 2026, and limits the sample to those resident in Norway in 2017. 2017 Residence is determined using the shareholder register. Migration dates are determined using the shareholder register, LinkedIn data, and newspaper articles. The dashed vertical is placed in April 2022 when the government confirmed that tighter exit tax rules were being developed. The shaded area indicates ambiguous migration dates (e.g., a range of dates from newspaper articles).

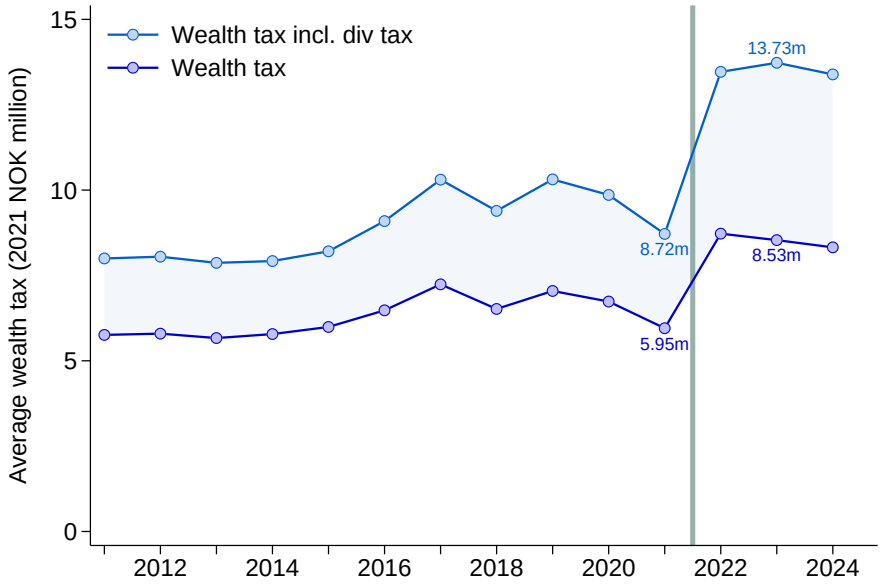
Figure OA.4: Inventor Mobility



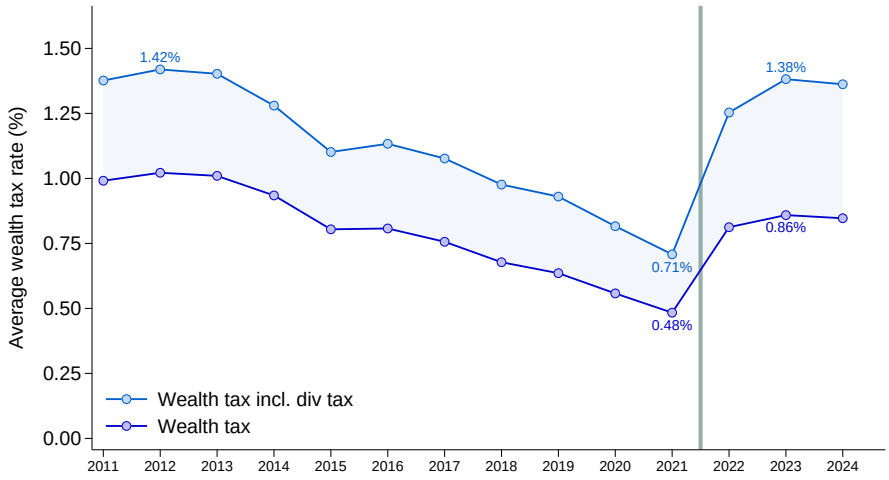
Notes: This figure shows migration rates of inventors for the Nordic countries. Panel A plots the share of inventors residing in a given country as of 2017 that subsequently move to another country. Residence and mobility is determined using the location of the most-recent employer on an inventor's LinkedIn profile at any point in time. Inventors are those who have filed patents prior to 2017. Panel C plots *inflows*: the number of arriving inventors, divided by the stock of inventors in the destination country as of 2017. These analyses rely on LinkedIn data from Revelio Labs linked with Patenting data provided by Orbis/Moody's. Panels B and D follow the specification in Figure 9.

FIGURE OA.5: WEALTH TAX EXPOSURE OVER TIME FOR THE TOP 0.01%: AVERAGE WEALTH TAX AND AVERAGE WEALTH TAX RATE

Panel A: Wealth Tax Amounts

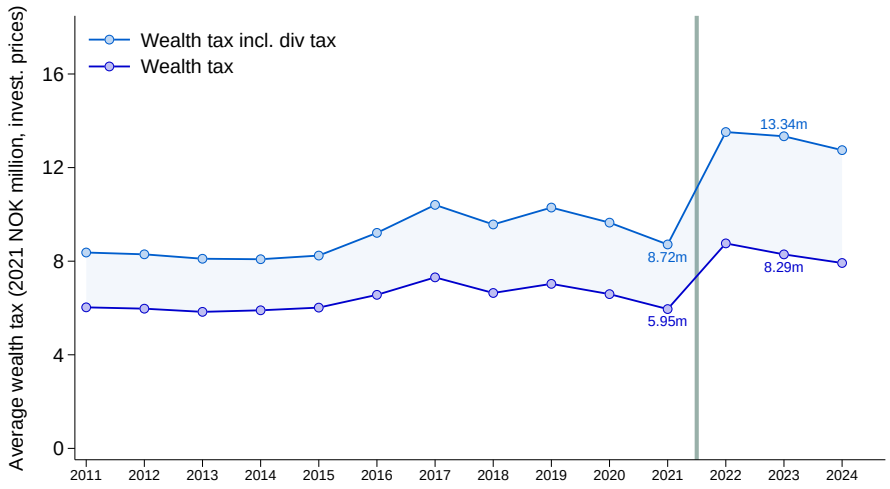


Panel B: Average Wealth Tax Rates



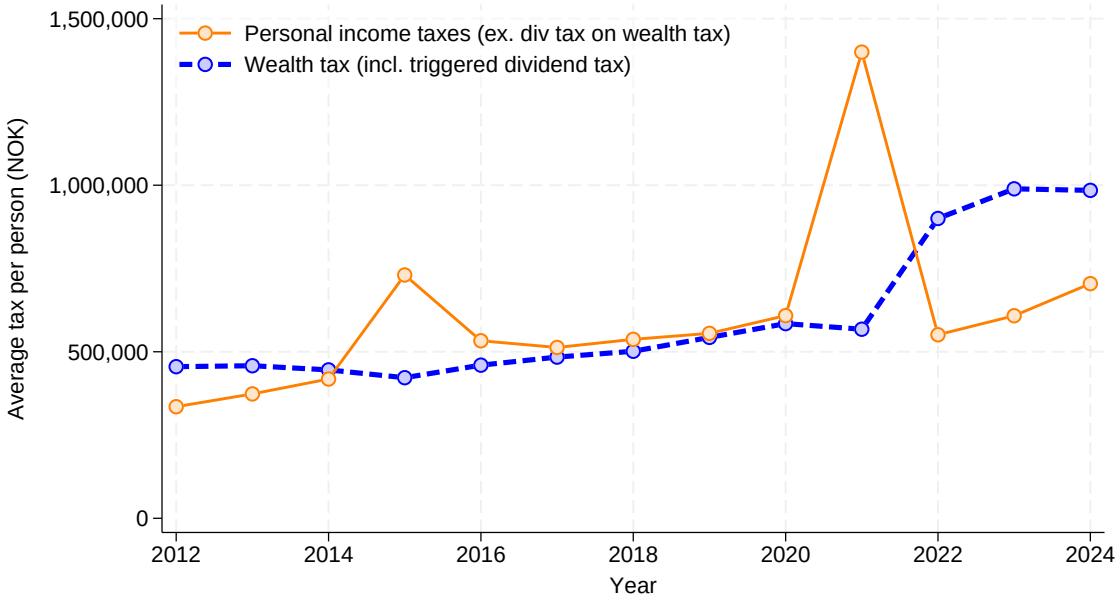
Notes: This figure shows measures of wealth tax exposure over time for those in the top 0.01% of taxable wealth in 2017. Panel A considers inflation-adjusted wealth tax amounts. Panel B considers the average wealth tax rates constructed as wealth tax amounts divided by undiscounted taxable net wealth (i.e., a measure of taxable net wealth in which assessment discounts on stocks and housing are removed). The underlying statistics are produced with microdata.no.

Figure OA.6: Wealth Tax Exposure Over Time for the Top 0.01%: Deflated using investment price index



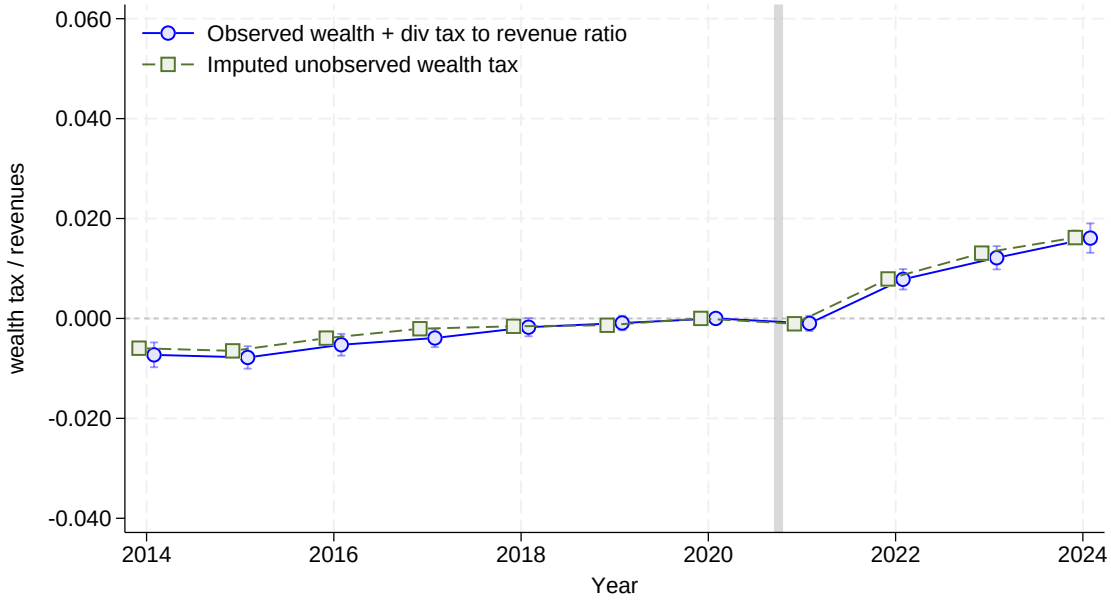
Notes: This figure shows inflation-adjusted wealth-tax amounts over time for those in the top 0.01% of taxable wealth in 2017. For this figure, I use the manufacturing investment price index (SSB Table 09181 A38 manufacturing aggregate) instead of the CPI. The underlying statistics are produced with microdata.no. See in-text figure 6 for the CPI-deflated amounts.

Figure OA.7: Wealth and Income Tax for Business Owners in Top-100 Lists



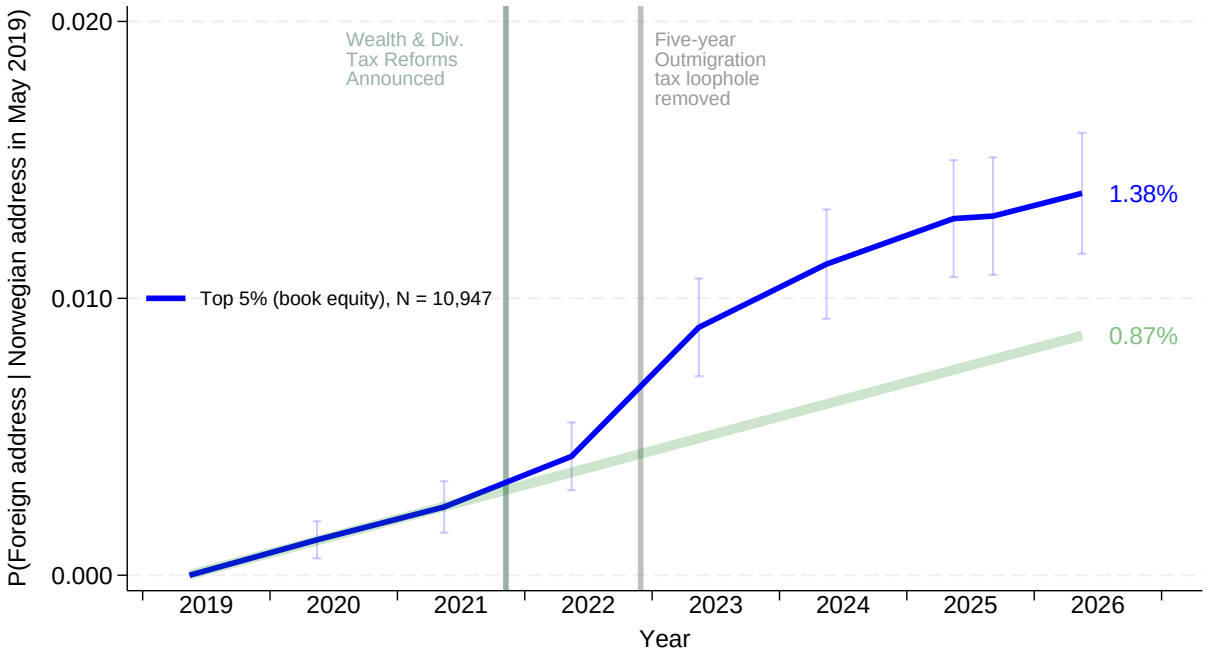
Notes: The solid orange line shows average total tax less calculated wealth tax and implied dividend tax. The dashed blue line shows average calculated wealth tax including implied dividend tax. Wealth taxes are calculated from reported taxable wealth and each year's tax schedule. Spikes in the remainder in 2015 and 2021 are consistent with anticipatory dividend payments before dividend tax increases during 2016–2019 and 2022–2023.

Figure OA.8: Effect on Wealth Tax Exposure
Ratio of Apportioned Wealth Tax to Revenue
Contrasting Actual and Imputed Amounts



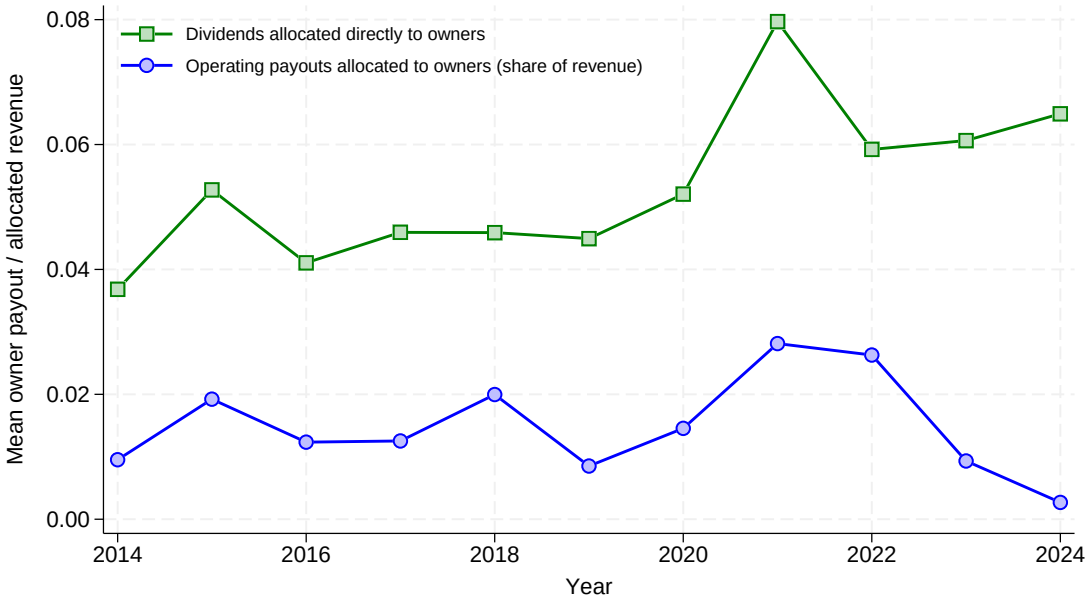
Notes: The figure compares taxes calculated from reported taxable wealth with the measure in main-text Figure 13, which also uses imputed wealth when reported wealth is unavailable. Both measures allocate taxes to treated firms and include implied dividend taxes. The reported-wealth-only measure additionally requires that owners with reported taxable wealth hold more than 50% of treated firms' shares.

Figure OA.9: Business Owner Outmigration in the Shareholder Register: Focusing on the Top 5%



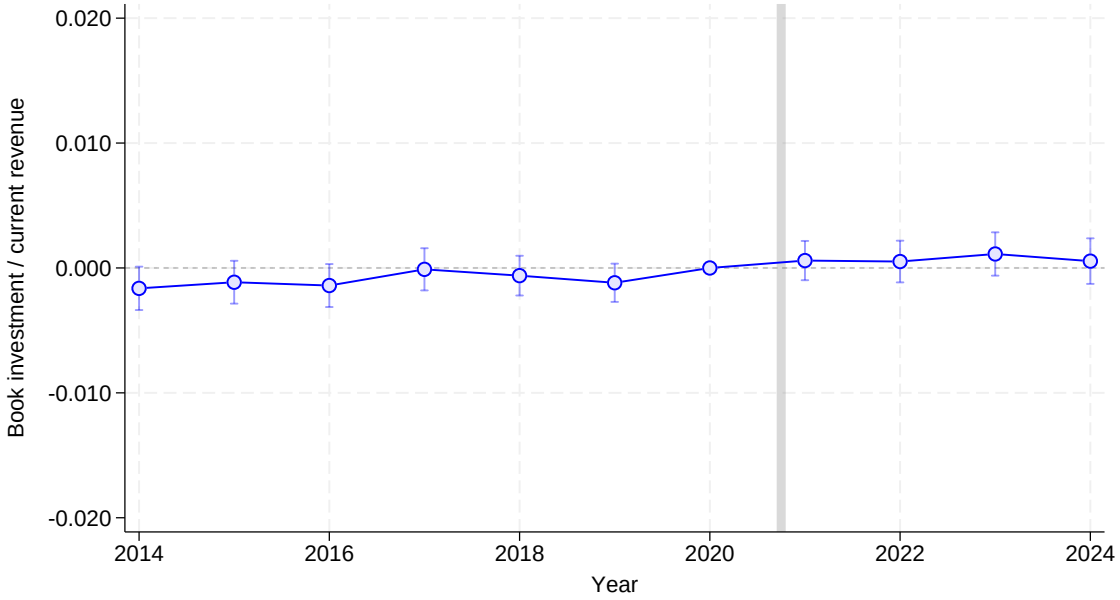
Notes: This figure shows outmigration in the top 5% instead of in the Top 1% (Figure 10). The purpose for widening the top bin is to examine whether I find the same implied number of individuals outmigrating. See Figure 10 for description of the methodology.

Figure OA.10: Dividends from in-sample versus directly-held firms



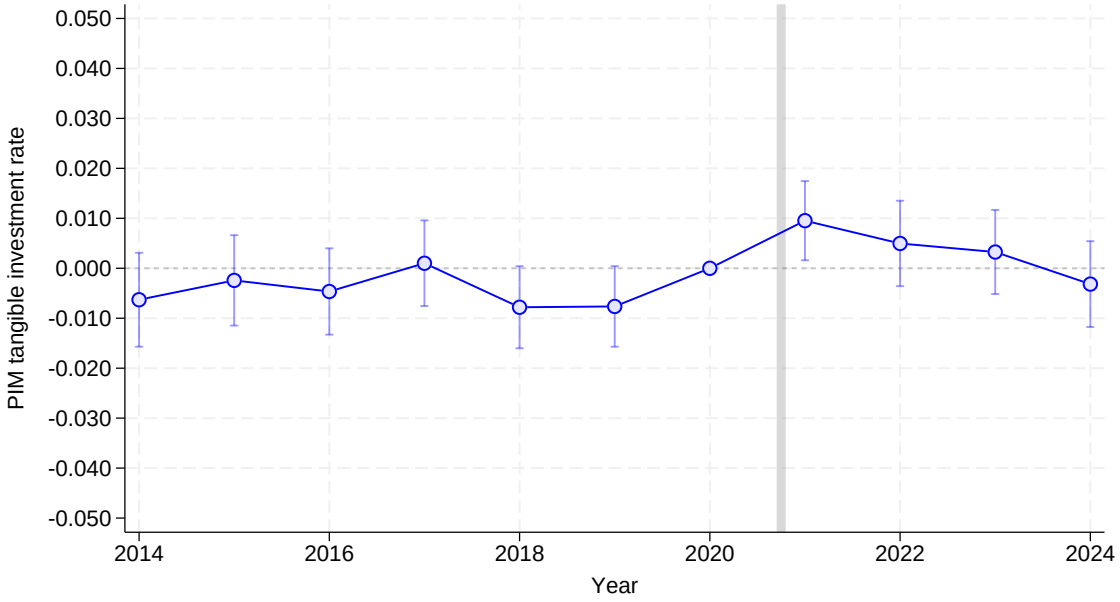
Notes: This figure contrasts operating dividends in my firm-outcomes analysis sample with dividends received directly by the same owners. All numbers are means, not regression coefficients. (i) The green line (squares) provides the ratio of directly-allocated dividends to revenues for the owners of treated firms. Dividends are allocated from all directly-held firms, even those not in my sample (e.g., those with no history of operating revenues). These dividends are received by the owners as individuals. They are calculated using the assumed accounting identity that dividends equal net income minus the increase in retained earnings. Ratios are bounded to lie in [-1,1] at the owner level prior to taking means. (ii) The blue line (circles) instead considers the ratio of allocated operating payouts (defined as in Figure 14) to revenues. These payouts are restricted to payouts from firms in my analysis sample. These dividends need not be received by any individuals due to indirect ownership. Both lines use total revenues allocated (directly and indirectly) from any firm (even if not in analysis sample) to owners in the denominator. This differs from, e.g., Figure 14, which considers the ratio of operating payouts to revenues at the firm level.

Figure OA.11: No Discernible Effect on Investment: As a Fraction of Revenue



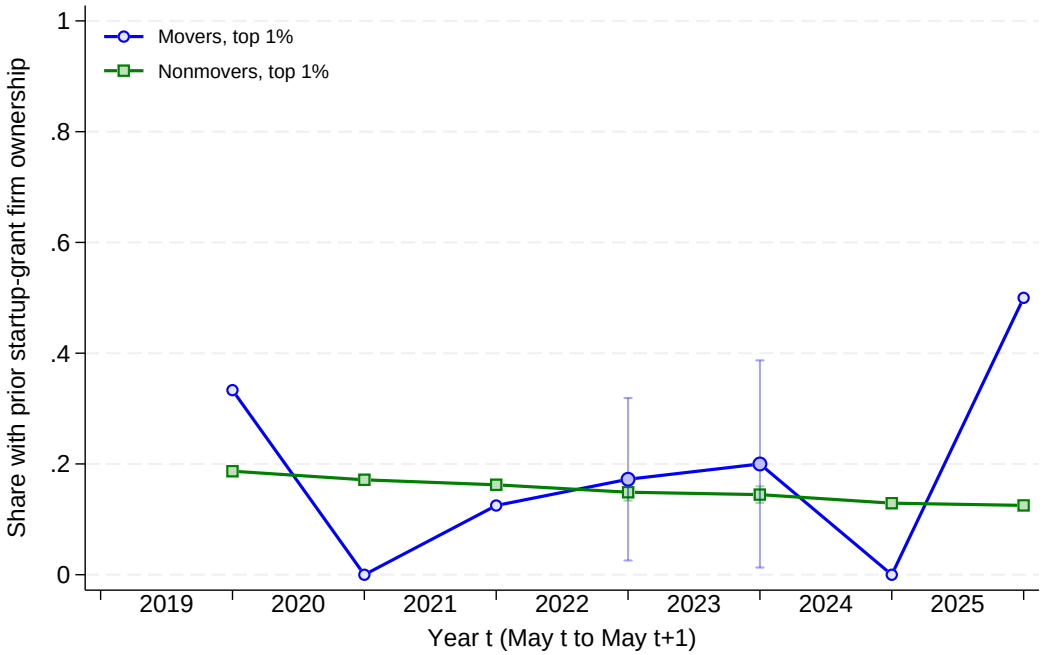
Notes: Instead of dividing investment by the PIM capital stock (see Figure 15), I now divide by concurrent revenues. The ratio is winsorized at the 5% and 95% levels.

Figure OA.12: No Discernible Effect on Investment: Excluding Outmigrants' Firms



Notes: This figure repeats Figure 15, but excludes the 373 treated firms whose largest ultimate owner in 2018 resided abroad in May 2024 according to the shareholder register. Owners who leave the shareholder register keep their last observed residence.

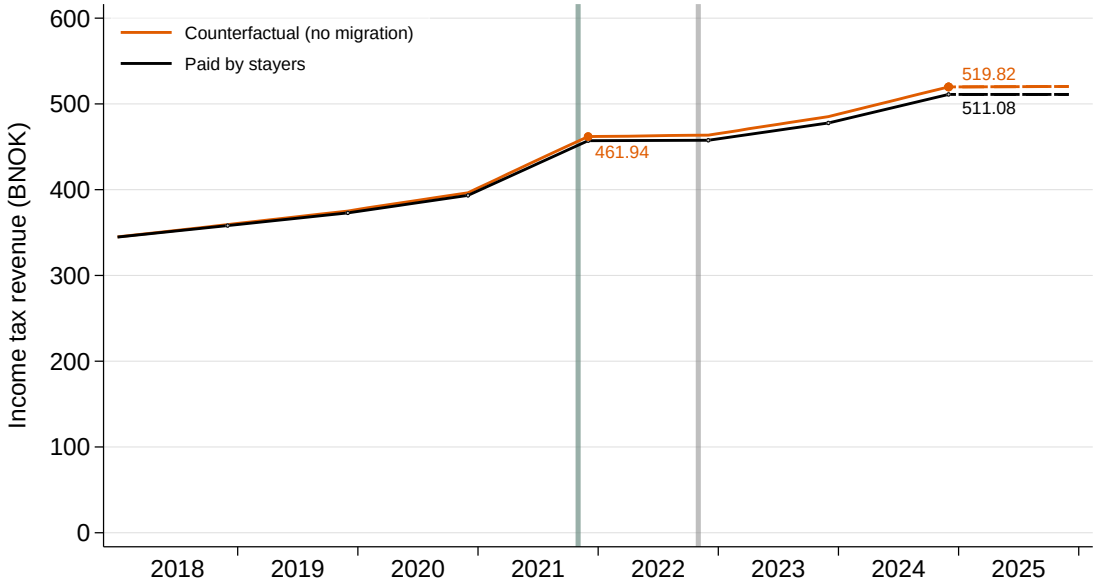
Figure OA.13: Whether Outmigrants Were Startup Investors



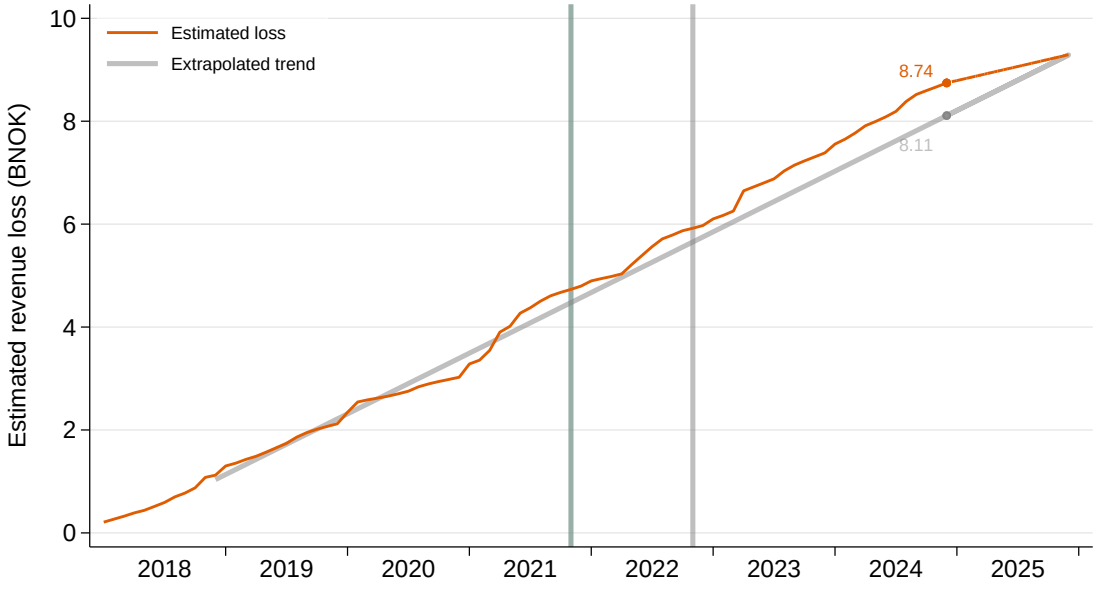
Notes: The figure considers the probability that outmigrants in different years were investors in grant-receiving startups. The answer is yes if an individual during any of the preceding four years was invested in a firm receiving a startup grant from Innovation Norway (and the startup grant must be received in the year of ownership or the year before). Iterated ownership shares must be at least 1%. 95% confidence intervals are based on robust standard errors from a linear probability model estimated via OLS. All CIs for the movers include the point-estimates for the nonmovers. See Figure 10 for the cumulative outmigration plot.

Figure OA.14: Income Tax Revenues Around The Reform, Using Administrative Data

Panel A: Income Tax Revenues With and Without Migration



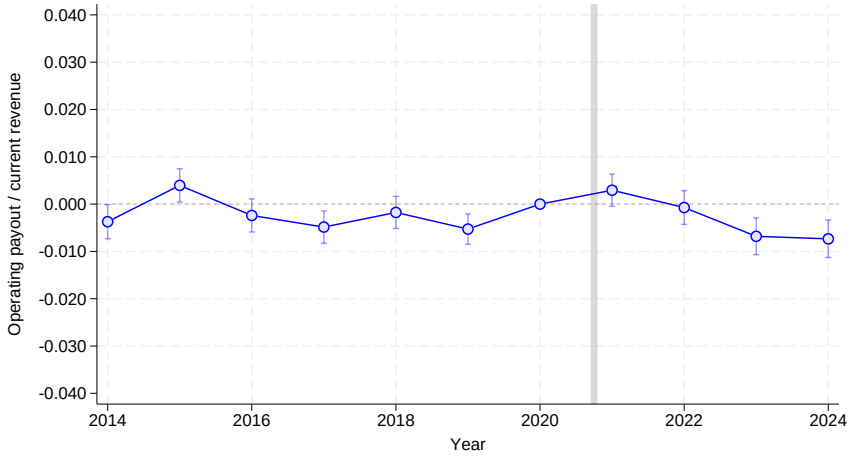
Panel B: Estimated Reduction in Income Tax Revenues



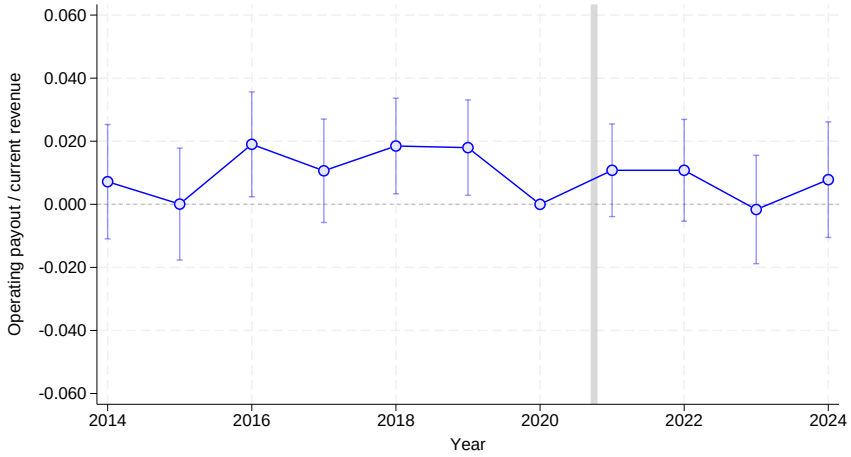
Notes: This figure shows income tax revenues over time. Relative to main-text Figure 5, I now consider revenues from personal income taxation as opposed to wealth taxation. Panel A, solid black line, shows aggregate income tax revenues paid by residents. The orange line adds the counterfactual payments of those who were resident on January 1st 2018 but subsequently emigrated. I assume that their year t income tax payment equals their 2017 tax payment scaled up 2017-wealth-group-specific cumulative growth rate (calculated on stayers). I calculate annualized tax payments at the monthly level by interpolating between year ends. I assign exact migration dates using data on address moves during the year of migration. Panel B, top orange line, shows the estimated revenue loss, which is the difference between the counterfactual revenues and the revenues paid by residents. The dark gray line is an extrapolated trend. The circles and numerical values are the numbers as of year-end 2024. Post 2024 numbers are based on an assumed zero growth of tax payments, since I do not observe end-of-year 2025 tax payments nor within-year move dates for 2025. The underlying data comes from microdata.no.

FIGURE OA.16: NO *INCREASE* IN PAYOUTS: Across Settings

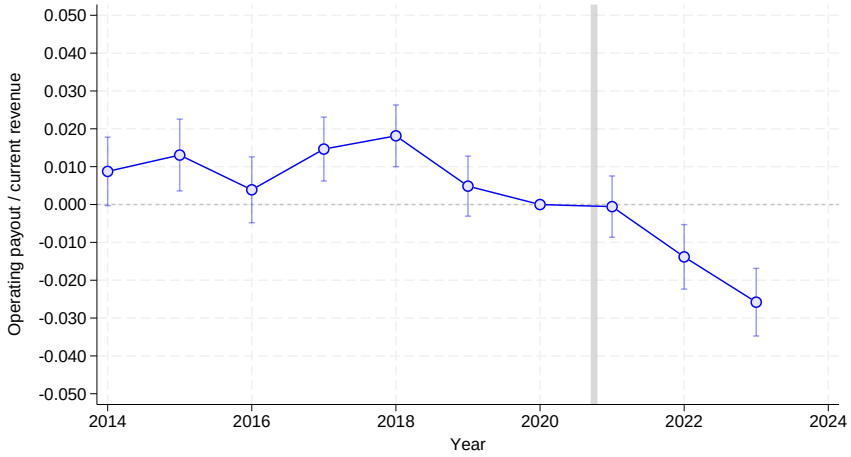
Panel A: Main within-Norway analysis



Panel B: Foreign sample



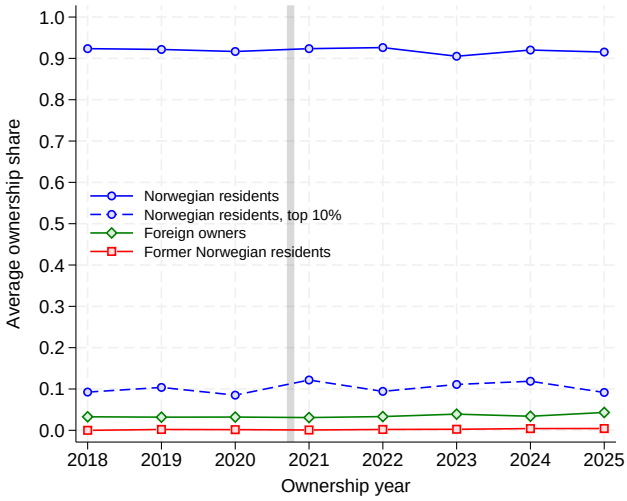
Panel C: Nordic cross-country analyses



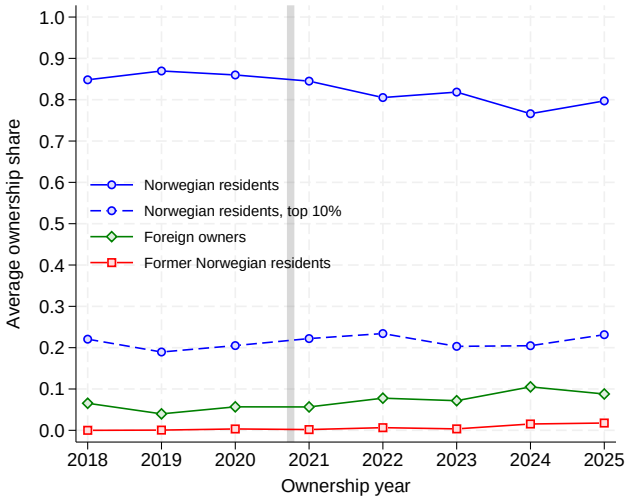
Notes: Operating dividends equal net income minus financial income and the change in retained earnings. These dividends are typically paid out in $t+1$. Operating dividends are divided by concurrent revenues and then winsorized at the 5% and 95% levels. This measure avoids counting dividends received and passed on through several ownership layers (see text for additional discussion). See main-text Figure 14. Panel C ends one year earlier than the other panels since most of my control firms are Swedish and Sweden has different accounting rules which change the timing relationship between changes in balance-sheet retained earnings and payouts.

Figure OA.17: Ownership Composition of Innovative Startups: Unweighted

(A) Startup Grantees

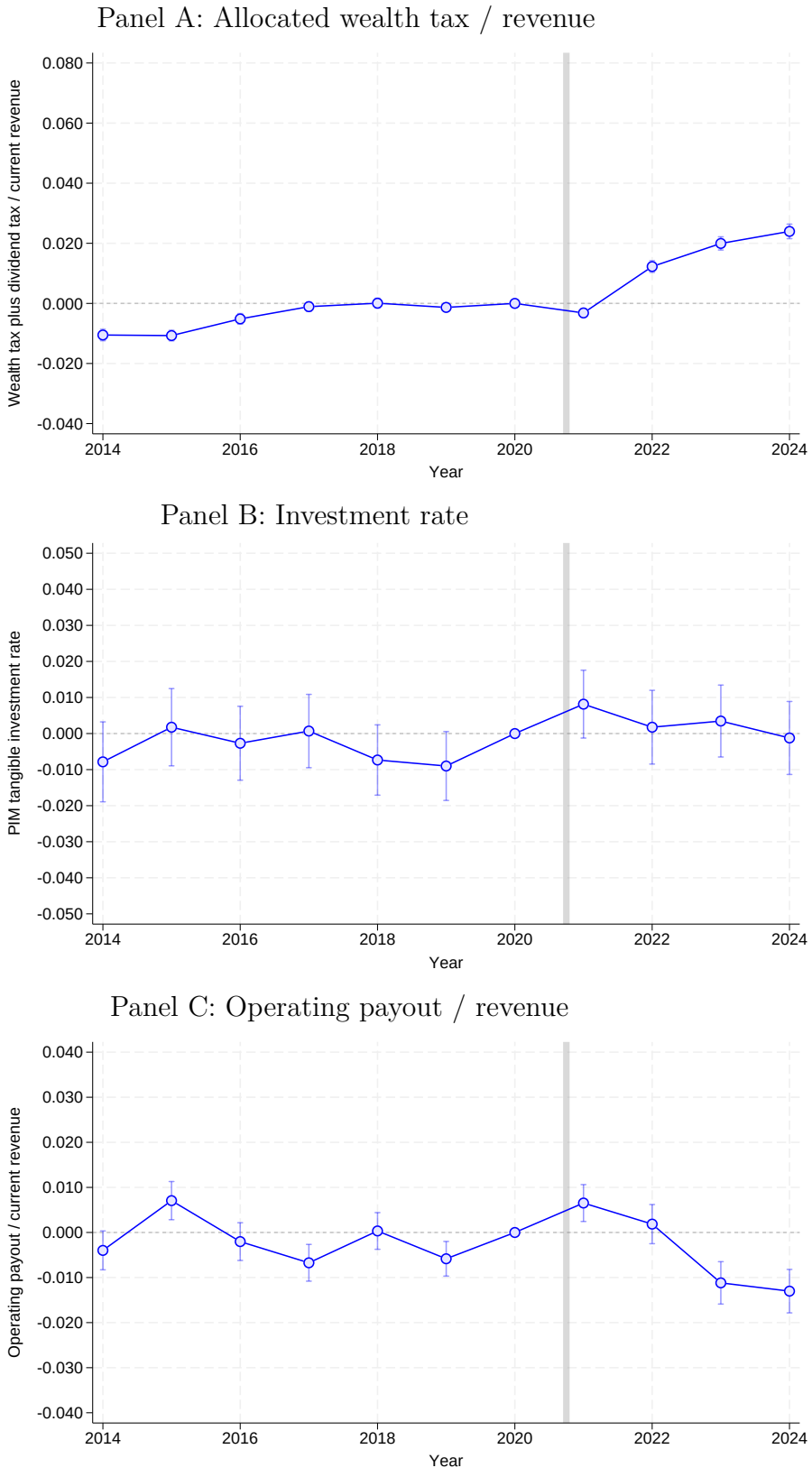


(B) Young Firms with Approved R&D Projects



Notes: This figure considers the ownership composition of two sets of startups over time. Relative to main-text Figure 18, this figure does not weight ownership shares by paid-in capital.

FIGURE OA.15: TREATED SUBSAMPLE WITH HIGHER WEALTH TAX EXPOSURE



Notes: All panels use the domestic sample. Treated firms are retained if the ratio of summed allocated wealth tax to summed revenues during 2016–2018 is above the median, requiring observations in all three years. All control firms are retained. Panel A considers wealth tax exposure. Panel B considers the investment rate. Panel C considers the payout rate.